

Recursive Macroeconomic Theory

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Complete Solutions

Chapter 2 — Time Series

All thirty end-of-chapter exercises, 2.1–2.30 (pp. 85–104)

Every exercise carries the full question text, a step-by-step solution with no skipped algebra, and a verification. Numerical answers are reproduced by the companion notebook `Resolucao/ljungqvist_sargent/ls_ch02.ipynb`, which has one section per exercise and checks every number against an independent route (closed form against recursion, truncated sum, simulation, a Riccati solver, or brute-force Gaussian conditioning), aborting on any mismatch. Where the book asks for Matlab (`doublej`, `bigshow3`, `kfilter`) the notebook does the same computation in Python.

Exercises are grouped by **topic**, not by number, following the order of the chapter's own sections. Blue boxes are odd-numbered exercises, purple boxes even.

Ljungqvist–Sargent is supplementary reading and lies outside the course syllabus; this file is for depth, not for the exam.

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Chapter 2 Time Series

The chapter builds two workhorse representations of a stochastic process and the tools that go with them.

Finite Markov chains (P, π_0) . Distributions move by $\pi_{t+1} = P'\pi_t$; a stationary distribution solves $(I - P')\pi = 0$ (a left unit eigenvector); an invariant function solves $(P - I)\bar{y} = 0$ (a right unit eigenvector, (2.2.13)). The chain is ergodic when every invariant function is constant on the support of π , and then time averages converge to population means (Theorem 2.2.5). Likelihoods factor as $\pi_{0,i_0} \prod_t P_{i_{t-1}i_t}$.

Linear state-space systems $x_{t+1} = Ax_t + Cw_{t+1}$, $y_t = Gx_t$. Means obey $\mu_{t+1} = A\mu_t$, covariances $\Sigma_{t+1} = A\Sigma_t A' + CC'$ with stationary fixed point (2.4.10); forecasts are $E_t x_{t+j} = A^j x_t$ and $E_t \sum \beta^j y_{t+j} = G(I - \beta A)^{-1} x_t$ (2.4.18); geometric sums of quadratic forms solve the Lyapunov pair (2.4.20). The spectrum (2.11.2) is the Fourier transform of the autocovariances.

Hidden states. When $y_t = Gx_t + v_t$ only is seen, the Kalman filter (2.7.14) is recursive population regression; its time-invariant version gives the innovations representation (2.9.1) and the VAR (2.9.3).

Control. Adding $u_t = -Fx_t$ turns (2.4.25) into policy evaluation and (2.4.30) into Howard policy improvement; the LQ permanent income model (2.12.18) is the running application.

Finite Markov Chains

Exercise 2.1 Likelihood of Three Histories

Question

Consider the Markov chain $(P, \pi_0) = \left(\begin{bmatrix} .9 & .1 \\ .3 & .7 \end{bmatrix}, \begin{bmatrix} .5 \\ .5 \end{bmatrix} \right)$, and a random variable $y_t = \bar{y}x_t$ where $\bar{y} = \begin{bmatrix} 1 & 5 \end{bmatrix}$. Compute the likelihood of the following three histories for y_t for $t = 0, 1, \dots, 4$:

(a) 1, 5, 1, 5, 1. (b) 1, 1, 1, 1, 1. (c) 5, 5, 5, 5, 5.

Solution:

Because $\bar{y}$ has distinct entries, each value of y_t identifies the state ($y = 1 \Leftrightarrow e_1$, $y = 5 \Leftrightarrow e_2$), so the likelihood of $y_0, \dots, y_4$ is the probability of the state path:

$$f(y_4, \dots, y_0) = \pi_{0,i_0} P_{i_0 i_1} P_{i_1 i_2} P_{i_2 i_3} P_{i_3 i_4}.$$

(a) Path 1, 2, 1, 2, 1: $.5 \times P_{12}P_{21}P_{12}P_{21} = .5 \times .1 \times .3 \times .1 \times .3 = .5 \times .0009 = \boxed{.00045}$.

(b) Path 1, 1, 1, 1, 1: $.5 \times P_{11}^4 = .5 \times .9^4 = .5 \times .6561 = \boxed{.32805}$.

(c) Path 2, 2, 2, 2, 2: $.5 \times P_{22}^4 = .5 \times .7^4 = .5 \times .2401 = \boxed{.12005}$.

Reading. The chain is persistent (large diagonal), so constant histories are far more likely than alternating ones; state 1 is stickier than state 2 ($.9 > .7$), so (b) beats (c).

Verification. `ls_ch02.ipynb`, §2.1 recomputes all three with the forward recursion $\alpha'_t = (\alpha'_{t-1}P) \odot \mathbf{1}\{y_t\}$. ✓

Exercise 2.2 Recovering P from Conditional Moments

Question

Consider a two-state Markov chain. Consider a random variable $y_t = \bar{y}x_t$ where $\bar{y} = \begin{bmatrix} 1 \\ 5 \end{bmatrix}$. It is known that $E(y_{t+1}|x_t) = \begin{bmatrix} 1.8 \\ 3.4 \end{bmatrix}$ and that $E(y_{t+1}^2|x_t) = \begin{bmatrix} 5.8 \\ 15.4 \end{bmatrix}$. Find a transition matrix consistent with these conditional expectations. Is this transition matrix unique (i.e., can you find another one that is consistent with these conditional expectations)?

Solution:

Conditional on $x_t = e_i$, $E(y_{t+1}|x_t) = \sum_j P_{ij}\bar{y}_j$, i.e. the vector of conditional means is $P\bar{y}$; likewise the conditional second moments are $P\bar{y}^2$ with $\bar{y}^2 = [1, 25]'$. Row i has two unknowns and satisfies

$$P_{i1} + P_{i2} = 1, \quad P_{i1} + 5P_{i2} = m_i \Rightarrow 1 + 4P_{i2} = m_i \Rightarrow P_{i2} = \frac{m_i - 1}{4}.$$

Row 1: $P_{12} = \frac{1.8-1}{4} = .2$, $P_{11} = .8$. Row 2: $P_{22} = \frac{3.4-1}{4} = .6$, $P_{21} = .4$.

$$P = \begin{bmatrix} .8 & .2 \\ .4 & .6 \end{bmatrix}$$

Check the second moments: row 1: $.8(1) + .2(25) = .8 + 5 = 5.8$ ✓; row 2: $.4(1) + .6(25) = .4 + 15 = 15.4$ ✓.

Uniqueness. Yes, P is unique. For each row the two equations (adding-up and the first moment) have coefficient matrix $\begin{bmatrix} 1 & 1 \\ 1 & 5 \end{bmatrix}$, with determinant $4 \neq 0$, so they already pin the row down. The second-moment restriction is an *overidentifying* restriction: it could have been violated, and it is not.

Verification. `ls_ch02.ipynb`, §2.2 stacks all six equations in the four unknowns: rank 4, zero residual, and rank 4 still holds without the second-moment rows. ✓

Exercise 2.3 Discounted Utility and Bayesian Model Choice

Question

Consumption is governed by an n -state Markov chain P, π_0 where P is a stochastic matrix and π_0 is an initial probability distribution. Consumption takes one of the values in the $n \times 1$ vector $\bar{c}$. A consumer ranks stochastic processes of consumption $t = 0, 1, \dots$ according to $E \sum_{t=0}^{\infty} \beta^t u(c_t)$ where E is the mathematical expectation and $u(c) = \frac{c^{1-\gamma}}{1-\gamma}$ for some parameter $\gamma \geq 1$. Let $u_i = u(\bar{c}_i)$. Let $v_i = E[\sum_{t=0}^{\infty} \beta^t u(c_t) | x_0 = \bar{c}_i]$ and $V = Ev$, where $\beta \in (0, 1)$ is a discount factor.

(a) Let u and v be the $n \times 1$ vectors whose i th components are u_i and v_i , respectively. Verify the following formulas for v and V : $v = (I - \beta P)^{-1}u$, and $V = \sum_i \pi_{0,i} v_i$.

(b) Consider the following two Markov processes: Process 1: $\pi_0 = \begin{bmatrix} .5 \\ .5 \end{bmatrix}$, $P = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$.

Process 2: $\pi_0 = \begin{bmatrix} .5 \\ .5 \end{bmatrix}$, $P = \begin{bmatrix} .5 & .5 \\ .5 & .5 \end{bmatrix}$. For both Markov processes, $\bar{c} = \begin{bmatrix} 1 \\ 5 \end{bmatrix}$. Assume that $\gamma = 2.5$, $\beta = .95$. Compute the unconditional discounted expected utility V for each of these processes. Which of the two processes does the consumer prefer? Redo the calculations for $\gamma = 4$. Now which process does the consumer prefer?

(c) An econometrician observes a sample of 10 observations of consumption rates for our consumer. He knows that one of the two preceding Markov processes generates the data, but he does not know which one. He assigns equal “prior probability” to the two chains. Suppose that the 10 successive observations on consumption are as follows: 1, 1, 1, 1, 1, 1, 1, 1, 1, 1. Compute the likelihood of this sample under process 1 and under process 2. Denote the likelihood function $\text{Prob}(\text{data} | \text{Model}_i)$, $i = 1, 2$.

(d) Suppose that the econometrician uses Bayes’ law to revise his initial probability estimates for the two models, where in this context Bayes’ law states: $\text{Prob}(M_i | \text{data}) = \frac{\text{Prob}(\text{data} | M_i) \cdot \text{Prob}(M_i)}{\sum_j \text{Prob}(\text{data} | M_j) \cdot \text{Prob}(M_j)}$ where M_i denotes model i . The denominator of this expression is the unconditional probability of the data. After observing the data sample, what probabilities does the econometrician place on the two possible models?

(e) Repeat the calculation in part d, but now assume that the data sample is 1, 5, 5, 1, 5, 5, 1, 5, 1, 5.

Solution:

(a) Since $E[u(c_t) | x_0 = e_i] = \sum_j (P^t)_{ij} u_j = (P^t u)_i$,

$$v = \sum_{t=0}^{\infty} \beta^t P^t u = (I - \beta P)^{-1} u,$$

where the geometric series converges because every eigenvalue of the stochastic matrix P has modulus ≤ 1 , so the spectral radius of βP is $\beta < 1$. Equivalently, splitting off $t = 0$ and using the Markov property, $v_i = u_i + \beta \sum_j P_{ij} v_j$, i.e. $v = u + \beta P v$, i.e. $(I - \beta P)v = u$. Finally, by the law of iterated expectations over the initial state, $V = E v(x_0) = \sum_i \pi_{0,i} v_i$. ✓

(b) *Process 1* ($P = I$): $v = (1 - \beta)^{-1}u$, so $V_1 = \frac{.5u_1 + .5u_2}{1 - \beta}$.

Process 2 ($P = \frac{1}{2}\mathbf{1}\mathbf{1}'$): $P^t u = \bar{u}\mathbf{1}$ for $t \geq 1$, with $\bar{u} \equiv .5u_1 + .5u_2$. Hence $v = u + \frac{\beta}{1-\beta}\bar{u}\mathbf{1}$ and

$$V_2 = \bar{u} + \frac{\beta}{1-\beta}\bar{u} = \frac{\bar{u}}{1-\beta} = V_1.$$

The two values are **identical for every** u , hence for every γ . Numbers:

$\gamma = 2.5$: $u(1) = \frac{1}{-1.5} = -.666667$, $u(5) = \frac{5^{-1.5}}{-1.5} = \frac{.0894427}{-1.5} = -.0596285$; $\bar{u} = -.363148$; $V_1 = V_2 = \frac{-.363148}{.05} = \boxed{-7.262951}$.

$\gamma = 4$: $u(1) = -\frac{1}{3}$, $u(5) = \frac{5^{-3}}{-3} = -.0026667$; $\bar{u} = -.168$; $V_1 = V_2 = \frac{-.168}{.05} = \boxed{-3.36}$.

The consumer is **indifferent** in both cases.

Reading. Both chains start from $[.5, .5]$, which is stationary for both, so in every period c_t equals 1 or 5 with probability one half under either process. Time-separable expected utility only sees these period-by-period marginals, so it is blind to the serial correlation: a single coin flip at $t = 0$ fixing consumption for life (process 1) is valued exactly like a fresh flip every period (process 2). Raising γ changes the level of V , never the ranking.

(c) The observations identify the states. Process 1: $\pi_{0,1} \cdot 1^9 = \boxed{.5}$. Process 2: $.5 \times .5^9 = .5^{10} = \frac{1}{1024} = \boxed{.000977}$.

(d) With equal priors the priors cancel:

$$\text{Prob}(M_1|\text{data}) = \frac{.5}{.5 + .5^{10}} = \frac{512}{513} = \boxed{.998051}, \quad \text{Prob}(M_2|\text{data}) = \boxed{.001949}.$$

(e) Process 1 has $P_{12} = 0$: a switch from 1 to 5 has probability zero, so $\text{Prob}(\text{data}|M_1) = 0$, while $\text{Prob}(\text{data}|M_2) = .5^{10}$. Hence $\boxed{\text{Prob}(M_1|\text{data}) = 0, \text{Prob}(M_2|\text{data}) = 1}$: a single switch refutes process 1.

Verification. `ls_ch02.ipynb`, §2.3: $(I - \beta P)^{-1}u$ against the truncated sum $\sum_{t \leq 3000} \beta^t P^t u$; the likelihoods by the forward recursion. ✓

Where this goes wrong

It is tempting to say the risk-averse consumer “must” dislike the fully persistent process 1 because it is riskier over a lifetime. With time-separable utility and no way to save, lifetime risk is irrelevant: only the distribution of c_t at each date enters V . Preferring early or late resolution of risk, or disliking persistence, requires non-separable (e.g. recursive) preferences.

Exercise 2.10 Convex Combinations of Stationary Distributions

Question

Let P be a transition matrix for a Markov chain. Suppose that P' has two distinct eigenvectors π_1, π_2 corresponding to unit eigenvalues of P' . Scale π_1 and π_2 so that they are vectors of probabilities (i.e., elements are nonnegative and sum to unity). Prove for any $\alpha \in [0, 1]$ that $\alpha\pi_1 + (1 - \alpha)\pi_2$ is an invariant distribution of P .

Solution:

Let $\pi_\alpha = \alpha\pi_1 + (1 - \alpha)\pi_2$. Three things must hold.

- (i) *Invariance.* By linearity and $P'\pi_k = \pi_k$: $P'\pi_\alpha = \alpha P'\pi_1 + (1 - \alpha)P'\pi_2 = \alpha\pi_1 + (1 - \alpha)\pi_2 = \pi_\alpha$.
- (ii) *Nonnegativity.* Each entry is $\alpha\pi_{1,i} + (1 - \alpha)\pi_{2,i} \geq 0$, a combination with nonnegative weights of nonnegative numbers.
- (iii) *Adding up.* $\mathbf{1}'\pi_\alpha = \alpha\mathbf{1}'\pi_1 + (1 - \alpha)\mathbf{1}'\pi_2 = \alpha + 1 - \alpha = 1$.

So π_α is a probability vector satisfying $\pi_\alpha = P'\pi_\alpha$, i.e. a stationary (invariant) distribution. ■ The set of stationary distributions is therefore convex, as the Remark after Definition 2.2.3 states; π_1, π_2 are its extreme points.

Verification. `ls_ch02.ipynb`, §2.10 checks five values of α for a block-diagonal chain. ✓

Exercise 2.11 Transient Distribution of an Absorbing Chain

Question

Consider a Markov chain with transition matrix $P = \begin{bmatrix} 1 & 0 & 0 \\ .2 & .5 & .3 \\ 0 & 0 & 1 \end{bmatrix}$ with initial distribution $\pi_0 = [\pi_{1,0} \ \pi_{2,0} \ \pi_{3,0}]'$. Let $\pi_t = [\pi_{1t} \ \pi_{2t} \ \pi_{3t}]'$ be the distribution over states at time t . Prove that for $t > 0$

$$\pi_{1t} = \pi_{1,0} + .2 \left(\frac{1 - .5^t}{1 - .5} \right) \pi_{2,0}, \quad \pi_{2t} = .5^t \pi_{2,0}, \quad \pi_{3t} = \pi_{3,0} + .3 \left(\frac{1 - .5^t}{1 - .5} \right) \pi_{2,0}.$$

Solution:

The law of motion $\pi_{t+1} = P'\pi_t$ reads, component by component,

$$\pi_{1,t+1} = \pi_{1t} + .2\pi_{2t}, \quad \pi_{2,t+1} = .5\pi_{2t}, \quad \pi_{3,t+1} = .3\pi_{2t} + \pi_{3t}.$$

State 2. The middle equation is a first-order linear difference equation: $\pi_{2t} = .5\pi_{2,t-1} = \dots = .5^t \pi_{2,0}$.

State 1. Summing the first equation from 0 to $t - 1$ and substituting $\pi_{2s} = .5^s \pi_{2,0}$:

$$\pi_{1t} = \pi_{1,0} + .2 \sum_{s=0}^{t-1} \pi_{2s} = \pi_{1,0} + .2 \pi_{2,0} \sum_{s=0}^{t-1} .5^s = \pi_{1,0} + .2 \left(\frac{1 - .5^t}{1 - .5} \right) \pi_{2,0}.$$

State 3. Identically, with .3 in place of .2. ■

As $t \rightarrow \infty$: $\pi_{1\infty} = \pi_{1,0} + .4\pi_{2,0}$, $\pi_{2\infty} = 0$, $\pi_{3\infty} = \pi_{3,0} + .6\pi_{2,0}$. The mass initially in the transient state 2 is split between the two absorbing states in the ratio .2 : .3.

Verification. 1s_ch02.ipynb, §2.11: iteration of P' against the formulas for $t = 1, \dots, 39$. ✓

Exercise 2.12 Columns of $(P')^t$

Question

Let P be a transition matrix for a Markov chain. For $t = 1, 2, \dots$, prove that the j th column of $(P')^t$ is the distribution across states at t when the initial distribution is $\pi_{j,0} = 1$, $\pi_{i,0} = 0 \forall i \neq j$.

Solution:

By the law of total probability and the Markov property, $\text{Prob}(x_{t+1} = e_i) = \sum_k \text{Prob}(x_t = e_k) P_{ki}$, i.e. $\pi_{t+1} = P' \pi_t$. By induction, $\pi_t = (P')^t \pi_0$: true at $t = 1$, and if $\pi_t = (P')^t \pi_0$ then $\pi_{t+1} = P' (P')^t \pi_0 = (P')^{t+1} \pi_0$. With $\pi_0 = e_j$, $\pi_t = (P')^t e_j$, and post-multiplying a matrix by e_j selects its j th column. ■ Equivalently: $((P')^t)_{ij} = (P^t)_{ji} = \text{Prob}(x_t = e_i | x_0 = e_j)$.

Verification. 1s_ch02.ipynb, §2.12: columns against iteration, and a 200,000-path simulation. ✓

Exercise 2.14 A Reducible Four-State Chain

Question

Consider a Markov chain with transition matrix $P = \begin{bmatrix} .5 & .5 & 0 & 0 \\ .1 & .9 & 0 & 0 \\ 0 & 0 & .9 & .1 \\ 0 & 0 & 0 & 1 \end{bmatrix}$ with state

space $X = \{e_i, i = 1, \dots, 4\}$ where e_i is the i th unit vector. A random variable y_t is a function $y_t = [1 \ 2 \ 3 \ 4]x_t$ of the underlying state.

- Find all stationary distributions of the Markov chain.
- Can you find a stationary distribution for which the Markov chain is ergodic?
- Compute all possible limiting values of the sample mean $\frac{1}{T} \sum_{t=0}^{T-1} y_t$ as $T \rightarrow \infty$.

Solution:

(a) Write $\pi = P'\pi$ component by component:

$$\pi_1 = .5\pi_1 + .1\pi_2, \quad \pi_2 = .5\pi_1 + .9\pi_2, \quad \pi_3 = .9\pi_3, \quad \pi_4 = .1\pi_3 + \pi_4.$$

The first gives $.5\pi_1 = .1\pi_2 \Rightarrow \pi_2 = 5\pi_1$ (the second says the same); the third gives $\pi_3 = 0$; the fourth then holds for any π_4 . With adding up, $6\pi_1 + \pi_4 = 1$. Setting $\alpha = 6\pi_1$:

$$\pi_\alpha = \alpha \begin{bmatrix} 1/6 \\ 5/6 \\ 0 \\ 0 \end{bmatrix} + (1 - \alpha) \begin{bmatrix} 0 \\ 0 \\ 0 \\ 1 \end{bmatrix}, \quad \alpha \in [0, 1].$$

There are two closed classes, $\{1, 2\}$ and $\{4\}$; state 3 is transient (it leaks into 4).

(b) Yes: $\alpha = 1$ or $\alpha = 0$ (the two extreme points). For $\alpha \in (0, 1)$ the function $\bar{y} = [1, 1, 0, 0]'$ satisfies $P\bar{y} = \bar{y}$, so it is invariant, yet it is not constant on the support $\{1, 2, 4\}$ of π_α : the chain is not ergodic.

(c) The limit depends on where the chain starts:

- $x_0 \in \{e_1, e_2\}$: the chain stays in $\{1, 2\}$, which is ergodic under $[1/6, 5/6]$, so by Theorem 2.2.5 the mean converges to $1 \cdot \frac{1}{6} + 2 \cdot \frac{5}{6} = \boxed{\frac{11}{6}}$.
- $x_0 \in \{e_3, e_4\}$: state 3 is left for 4 with probability one (after a geometric time with mean 10), and 4 is absorbing, so the mean converges to $\boxed{4}$.

Drawing x_0 from π_α , the limit is the random variable $E[y_\infty | x_0]$ of (2.2.11): $11/6$ with probability α and 4 with probability $1 - \alpha$.

Verification. `ls_ch02.ipynb`, §2.14: sample means over 200,000 periods from each starting state: 1.8341, 1.8327, 3.9999, 4.0000. ✓

Exercise 2.16 Law of Large Numbers for Indicators

Question

Consider an n -state Markov chain with state space $X = \{e_i, i = 1, \dots, n\}$ where e_i is the i th unit vector. Consider the indicator variable $I_{it} = e_i x_t$ which equals 1 if $x_t = e_i$ and 0 otherwise. Suppose that the chain has a unique stationary distribution and that it is ergodic. Let π be the stationary distribution.

- Verify that $E I_{it} = \pi_i$.
- Prove that $\frac{1}{T} \sum_{t=0}^{T-1} I_{it} = \pi_i$ as $T \rightarrow \infty$ with probability one with respect to the stationary distribution π .

Solution:

- I_{it} is Bernoulli, so $E I_{it} = \text{Prob}(x_t = e_i) = \left((P')^t \pi \right)_i = \pi_i$, because $x_0 \sim \pi$ and $P'\pi = \pi$ make the distribution of x_t equal to π at every t .

- (b) $I_{it} = \bar{y}'x_t$ with $\bar{y} = e_i$, a random variable defined on the stationary ergodic chain (P, π) . Theorem 2.2.5 gives $\frac{1}{T} \sum_{t=1}^T I_{it} \rightarrow E[I_{i0}] = \pi_i$ w.p. 1. The requested sum runs from 0 to $T-1$; the difference is $\frac{1}{T} \sum_{t=0}^{T-1} I_{it} - \frac{1}{T} \sum_{t=1}^T I_{it} = \frac{I_{i0} - I_{iT}}{T}$, bounded by $1/T \rightarrow 0$. ■

Reading. The long-run fraction of time spent in state i equals its stationary probability. That is what licenses estimating π by sample frequencies (used in Exercise 2.17).

Verification. `ls_ch02.ipynb`, § 2.16: frequencies over 300,000 periods match π within .005. ✓

Exercise 2.17 Lake Model

Question

A worker can be in one of two states, state 1 (unemployed) or state 2 (employed). At the beginning of each period, a previously unemployed worker has probability $\lambda = \int_{\bar{w}}^B dF(w)$ of becoming employed. Here $\bar{w}$ is his reservation wage and $F(w)$ is the c.d.f. of a wage offer distribution. We assume that $F(0) = 0, F(B) = 1$. At the beginning of each period an unemployed worker draws one and only one wage offer from F . Successive draws from F are i.i.d. The worker's decision rule is to accept the job if $w \geq \bar{w}$, and otherwise to reject it and remain unemployed one more period. Assume that $\bar{w}$ is such that $\lambda \in (0, 1)$. At the beginning of each period, a previously employed worker is fired with probability $\delta \in (0, 1)$. Newly fired workers must remain unemployed for one period before drawing a new wage offer.

- Let the state space be $X = \{e_i, i = 1, 2\}$ where e_i is the i th unit vector. Describe the Markov chain on X that is induced by the description above. Compute all stationary distributions of the chain. Under what stationary distributions, if any, is the chain ergodic?
- Suppose that $\lambda = .05, \delta = .25$. Compute a stationary distribution. Compute the fraction of his life that an infinitely lived worker would spend unemployed.
- Drawing the initial state from the stationary distribution, compute the joint distribution $g_{ij} = \text{Prob}(x_t = e_i, x_{t-1} = e_j)$ for $i = 1, 2, j = 1, 2$.
- Define an indicator function by letting $I_{ij,t} = 1$ if $x_t = e_i, x_{t-1} = e_j$ at time t , and 0 otherwise. Compute $\lim_{T \rightarrow \infty} \frac{1}{T} \sum_{t=1}^T I_{ij,t}$ for all four i, j combinations.
- Building on your results in part d, construct method of moments estimators of λ and δ . Assuming that you know the wage offer distribution F , construct a method of moments estimator of the reservation wage $\bar{w}$.
- Compute maximum likelihood estimators of λ and δ .
- Compare the estimators you derived in parts e and f.
- Extra credit.* Compute the asymptotic covariance matrix of the maximum likelihood estimators of λ and δ .

Solution:

- From unemployment the worker finds a job with probability λ ; from employment

he is fired with probability δ :

$$P = \begin{bmatrix} 1 - \lambda & \lambda \\ \delta & 1 - \delta \end{bmatrix}.$$

Stationarity, first component: $\pi_1 = (1 - \lambda)\pi_1 + \delta\pi_2 \Rightarrow \lambda\pi_1 = \delta\pi_2$ (inflows to unemployment equal outflows). With $\pi_1 + \pi_2 = 1$:

$$\pi = \left[\frac{\delta}{\lambda + \delta}, \frac{\lambda}{\lambda + \delta} \right]'$$

The eigenvalues of P are 1 and $1 - \lambda - \delta \in (-1, 1)$, so the unit eigenvalue is simple: π is the **unique** stationary distribution and the only invariant functions are constants. The chain is ergodic under π .

- (b) $\pi_1 = \frac{.25}{.30} = \boxed{.8333}$, $\pi_2 = \boxed{.1667}$. By Theorem 2.2.5 applied to I_{1t} (Exercise 2.16), a worker spends a fraction $\boxed{5/6}$ of his life unemployed.

- (c) $g_{ij} = \text{Prob}(x_{t-1} = e_j) \text{Prob}(x_t = e_i | x_{t-1} = e_j) = \pi_j P_{ji}$:

$$\begin{aligned} g_{11} &= \frac{5}{6}(.95) = .791667, & g_{21} &= \frac{5}{6}(.05) = .041667, \\ g_{12} &= \frac{1}{6}(.25) = .041667, & g_{22} &= \frac{1}{6}(.75) = .125. \end{aligned}$$

($g_{21} = g_{12}$ is the flow balance of part a.)

- (d) The pair (x_t, x_{t-1}) is itself a stationary Markov chain on four states, ergodic because x_t is; applying Theorem 2.2.5 to the indicator of each pair, $\frac{1}{T} \sum_{t=1}^T I_{ij,t} \rightarrow g_{ij}$ w.p. 1, with the numbers of part c.
- (e) Invert the population moments: $g_{21}/(g_{11} + g_{21}) = \pi_1 \lambda / \pi_1 = \lambda$ and $g_{12}/(g_{12} + g_{22}) = \delta$. Replacing g_{ij} by sample frequencies $\hat{g}_{ij} = \frac{1}{T} \sum_t I_{ij,t}$:

$$\hat{\lambda}_{MM} = \frac{\hat{g}_{21}}{\hat{g}_{11} + \hat{g}_{21}}, \quad \hat{\delta}_{MM} = \frac{\hat{g}_{12}}{\hat{g}_{12} + \hat{g}_{22}}, \quad \hat{w}_{MM} = F^{-1}(1 - \hat{\lambda}_{MM}),$$

the last from $\lambda = 1 - F(\bar{w})$ (with F continuous and strictly increasing near $\bar{w}$).

- (f) Let n_{ij} be the number of transitions from state i to state j . Conditional on x_0 , the log likelihood is

$$\ell = n_{11} \log(1 - \lambda) + n_{12} \log \lambda + n_{21} \log \delta + n_{22} \log(1 - \delta).$$

$$\partial \ell / \partial \lambda = -\frac{n_{11}}{1 - \lambda} + \frac{n_{12}}{\lambda} = 0 \Rightarrow \boxed{\hat{\lambda}_{ML} = \frac{n_{12}}{n_{11} + n_{12}}}; \text{ similarly } \boxed{\hat{\delta}_{ML} = \frac{n_{21}}{n_{21} + n_{22}}} \text{ (fraction of the unemployed who find jobs; of the employed who lose them).}$$

- (g) They are **identical**: $T\hat{g}_{21}$ counts pairs $(x_{t-1} = e_1, x_t = e_2)$, which is n_{12} , and $T(\hat{g}_{11} + \hat{g}_{21}) = n_{11} + n_{12}$; likewise for δ . By invariance of ML, $F^{-1}(1 - \hat{\lambda}_{ML})$ is also the MLE of $\bar{w}$. In the simulated sample of `ls_ch02.ipynb`, § 2.17 ($T = 200,000$) both give $\hat{\lambda} = .05021$, $\hat{\delta} = .24933$.
- (h) Second derivatives: $\partial^2 \ell / \partial \lambda^2 = -\frac{n_{11}}{(1-\lambda)^2} - \frac{n_{12}}{\lambda^2}$, cross derivative 0. Per observation, $En_{11}/T = \pi_1(1-\lambda)$ and $En_{12}/T = \pi_1\lambda$, so the information for λ is $\pi_1 \left[\frac{1-\lambda}{(1-\lambda)^2} + \frac{\lambda}{\lambda^2} \right] = \frac{\pi_1}{\lambda(1-\lambda)}$, and for δ , $\frac{\pi_2}{\delta(1-\delta)}$. Hence

$$\sqrt{T} \begin{bmatrix} \hat{\lambda} - \lambda \\ \hat{\delta} - \delta \end{bmatrix} \rightarrow \mathcal{N} \left(0, \begin{bmatrix} \frac{\lambda(1-\lambda)}{\pi_1} & 0 \\ 0 & \frac{\delta(1-\delta)}{\pi_2} \end{bmatrix} \right) = \mathcal{N} \left(0, \begin{bmatrix} \frac{\lambda(1-\lambda)(\lambda+\delta)}{\delta} & 0 \\ 0 & \frac{\delta(1-\delta)(\lambda+\delta)}{\lambda} \end{bmatrix} \right).$$

At $\lambda = .05, \delta = .25$: $\frac{.05(.95)}{.8333} = \boxed{.057}$ and $\frac{.25(.75)}{.1667} = \boxed{1.125}$. Each is a binomial variance divided by the expected share of observations that are informative about that parameter; δ is imprecise because workers are rarely employed.

Verification. `ls_ch02.ipynb`, § 2.17: 1,000 samples of $T = 5000$ give $T \cdot \text{Var} = (.0591, 1.168)$ and $T \cdot \text{Cov} \approx -.019$, against $(.057, 1.125, 0)$. ✓

Exercise 2.18 Random Walk

Question

A Markov chain has state space $X = \{e_i, i = 1, \dots, 4\}$ where e_i is the unit vector

and transition matrix $P = \begin{bmatrix} 1 & 0 & 0 & 0 \\ .5 & 0 & .5 & 0 \\ 0 & .5 & 0 & .5 \\ 0 & 0 & 0 & 1 \end{bmatrix}$. A random variable $y_t = \bar{y}x_t$ is defined

by $\bar{y} = [1 \ 2 \ 3 \ 4]$.

- Find all stationary distributions of this Markov chain.
- Under what stationary distributions, if any, is this chain ergodic? Compute invariant functions of P .
- Compute $E[y_{t+1}|x_t]$ for $x_t = e_i, i = 1, \dots, 4$.
- Compare your answer to part (c) with (2.2.12). Is $y_t = \bar{y}'x_t$ invariant? If not, what hypothesis of Theorem 2.2.4 is violated?
- The stochastic process $y_t = \bar{y}'x_t$ is evidently a bounded martingale. Verify that y_t converges almost surely to a constant. To what constant(s) does it converge?

Solution:

- $\pi = P'\pi$: $\pi_2 = .5\pi_3$ and $\pi_3 = .5\pi_2$ give $\pi_2 = .25\pi_2 \Rightarrow \pi_2 = \pi_3 = 0$; then $\pi_1 = \pi_1 + .5\pi_2$ and $\pi_4 = .5\pi_3 + \pi_4$ hold for any π_1, π_4 : $\pi = [\alpha, 0, 0, 1 - \alpha]'$, $\alpha \in [0, 1]$.
- Invariant functions solve $P\bar{y} = \bar{y}$: rows 1 and 4 are identities; rows 2–3 give

$\bar{y}_2 = \frac{\bar{y}_1 + \bar{y}_3}{2}$, $\bar{y}_3 = \frac{\bar{y}_2 + \bar{y}_4}{2}$, whose solution is

$$\bar{y}_2 = \frac{2\bar{y}_1 + \bar{y}_4}{3}, \quad \bar{y}_3 = \frac{\bar{y}_1 + 2\bar{y}_4}{3}, \quad \text{i.e.} \quad \bar{y} = \bar{y}_1 \begin{bmatrix} 1 \\ 2/3 \\ 1/3 \\ 0 \end{bmatrix} + \bar{y}_4 \begin{bmatrix} 0 \\ 1/3 \\ 2/3 \\ 1 \end{bmatrix}.$$

A two-dimensional space (the basis vectors are the probabilities of absorption at 1 and at 4). The chain is ergodic only for $\alpha \in \{0, 1\}$: for $\alpha \in (0, 1)$ the invariant function $[1, \frac{2}{3}, \frac{1}{3}, 0]'$ takes different values on the two support points e_1, e_4 .

- (c) $P\bar{y} = [1, .5(1) + .5(3), .5(2) + .5(4), 4]' = [1, 2, 3, 4]' = \bar{y}$.
- (d) Part c is exactly (2.2.12): $E[y_{t+1}|x_t] = y_t$, so y_t is a martingale and $\bar{y}$ is a right unit eigenvector. Under any stationary distribution $[\alpha, 0, 0, 1 - \alpha]'$ the chain sits in an absorbing state forever, so $y_t = y_0$ on every path of positive probability: y is invariant, as the theorem says. But started from e_2 or e_3 , y_t moves. This does not contradict Theorem 2.2.4, because its hypothesis that x_0 is drawn from a **stationary** distribution fails: any initial distribution that puts mass on e_2, e_3 is not stationary. The proof breaks at the step $Ey_{t+1}^2 = Ey_t^2$.
- (e) Each period spent in $\{2, 3\}$ ends in absorption with probability .5, so $\text{Prob}(\text{not absorbed by } t) = .5^t \rightarrow 0$: y_t is eventually constant almost surely. Let h_i be the probability of absorption at 4 starting from e_i : $h_1 = 0, h_4 = 1$, $h_2 = .5h_1 + .5h_3$, $h_3 = .5h_2 + .5h_4$, so $h_2 = \frac{1}{3}$, $h_3 = \frac{2}{3}$. Hence

$$x_0 = e_2 : y_t \rightarrow \begin{cases} 1 & \text{w.p. } 2/3 \\ 4 & \text{w.p. } 1/3 \end{cases}, \quad x_0 = e_3 : y_t \rightarrow \begin{cases} 1 & \text{w.p. } 1/3 \\ 4 & \text{w.p. } 2/3 \end{cases}.$$

The limit is a (random) constant, 1 or 4. Martingale check: $Ey_\infty = 1 \cdot \frac{2}{3} + 4 \cdot \frac{1}{3} = 2 = y_0$. ✓

Verification. `ls_ch02.ipynb`, §2.18: $(I - Q)^{-1}R$ with Q, R the transient/absorbing blocks gives h ; 100,000 simulated paths from e_2 are absorbed at 1 with frequency .6673.

✓

Linear Stochastic Difference Equations and Prediction

Question

Consider the univariate stochastic process $y_{t+1} = \alpha + \sum_{j=1}^4 \rho_j y_{t+1-j} + c w_{t+1}$ where w_{t+1} is a scalar martingale difference sequence adapted to $J_t = [w_t, \dots, w_1, y_0, y_{-1}, y_{-2}, y_{-3}]$, $\alpha = \mu(1 - \sum_j \rho_j)$ and the ρ_j 's are such that the matrix

$$A = \begin{bmatrix} \rho_1 & \rho_2 & \rho_3 & \rho_4 & \alpha \\ 1 & 0 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 1 \end{bmatrix} \text{ has all of its eigenvalues in modulus bounded below unity.}$$

- (a) Show how to map this process into a first-order linear stochastic difference equation.
- (b) For each of the following examples, if possible, assume that the initial conditions are such that y_t is covariance stationary. For each case, state the appropriate initial conditions. Then compute the covariance stationary mean and variance of y_t assuming the following parameter sets of parameter values: i. $\rho = [1.2 \ - .3 \ 0 \ 0]$, $\mu = 10$, $c = 1$. ii. $\rho = [1.2 \ - .3 \ 0 \ 0]$, $\mu = 10$, $c = 2$. iii. $\rho = [.9 \ 0 \ 0 \ 0]$, $\mu = 5$, $c = 1$. iv. $\rho = [.2 \ 0 \ 0 \ .5]$, $\mu = 5$, $c = 1$. v. $\rho = [.8 \ .3 \ 0 \ 0]$, $\mu = 5$, $c = 1$. *Hint 1:* The Matlab command `X=doublej(A,C*C')` computes the solution of the matrix equation $AXA' + CC' = X$. *Hint 2:* The mean vector is the eigenvector of A associated with a unit eigenvalue, scaled so that the mean of unity in the state vector is unity.
- (c) For each case in part b, compute the h_j 's in $E_t y_{t+5} = \gamma_0 + \sum_{j=0}^3 h_j y_{t-j}$.
- (d) For each case in part b, compute the $\tilde{h}_j$'s in $E_t \sum_{k=0}^{\infty} .95^k y_{t+k} = \sum_{j=0}^3 \tilde{h}_j y_{t-j}$.
- (e) For each case in part b, compute the autocovariance $E(y_t - \mu_y)(y_{t-k} - \mu_y)$ for the three values $k = 1, 5, 10$.

Solution:

- (a) Stack four lags and a constant:

$$x_t = \begin{bmatrix} y_t \\ y_{t-1} \\ y_{t-2} \\ y_{t-3} \\ 1 \end{bmatrix}, \quad x_{t+1} = Ax_t + Cw_{t+1}, \quad C = \begin{bmatrix} c \\ 0 \\ 0 \\ 0 \\ 0 \end{bmatrix}, \quad y_t = Gx_t, \quad G = [1 \ 0 \ 0 \ 0 \ 0].$$

Row 1 of A is the AR equation, rows 2–4 shift the lags down, row 5 reproduces the constant.

- (b) The eigenvalues of A are 1 (the constant) and the roots λ of $\lambda^4 - \rho_1 \lambda^3 - \rho_2 \lambda^2 - \rho_3 \lambda - \rho_4 = 0$. Covariance stationarity requires the latter inside the unit circle. *Mean:* $(I - A)\mu_x = 0$ with last entry 1; row 1 gives $\mu_y = \sum_j \rho_j \mu_y + \alpha \Rightarrow \mu_y = \frac{\alpha}{1 - \sum \rho_j} = \mu$, so $\mu_x = [\mu, \mu, \mu, \mu, 1]'$. *Initial conditions:* draw x_0 with mean μ_x and covariance Σ_∞ solving $\Sigma_\infty = A\Sigma_\infty A' + CC'$ (zero last row and column). Then $\text{Var}(y_t) = (\Sigma_\infty)_{11}$ for all t .

case	AR roots λ	μ_y	$\text{Var}(y)$
i	.8449, .3551	10	7.428571
ii	.8449, .3551	10	29.714286
iii	.9	5	5.263158
iv	.8957, $-.7950$, $.0497 \pm .8365i$	5	1.476398
v	1.0782, $-.2782$	—	not stationary

Closed-form checks. AR(2), $\text{Var}(y) = \frac{(1-\rho_2)c^2}{(1+\rho_2)[(1-\rho_2)^2-\rho_1^2]} = \frac{1.3}{.7(1.69-1.44)} = \frac{1.3}{.175} = 7.428571$ for $c = 1$, and $4\times$ that for $c = 2$. AR(1): $\frac{1}{1-.81} = 5.263158$. Case v: $\rho_1 + \rho_2 = 1.1 > 1$ puts a root at $\lambda = \frac{.8+\sqrt{.64+1.2}}{2} = 1.0782 > 1$. The mean fixed point μ_x still exists but is unstable, and Σ_t grows without bound for every x_0 because $c \neq 0$. No initial condition makes y_t covariance stationary.

(c) $E_t y_{t+5} = GA^5 x_t$ (2.4.17), so $h_j = (A^5)_{1,j+1}$ and $\gamma_0 = (A^5)_{15}$:

case	γ_0	h_0	h_1	h_2	h_3
i, ii	5.2156	.73872	-.26028	0	0
iii	2.04755	.59049	0	0	0
iv	2.6244	.20032	.02	.004	.2508
v	-2.4038	1.15808	.32268	0	0

The forecast does not depend on c (i and ii coincide). Case iii: $h_0 = .9^5$ and $\gamma_0 = 5(1 - .9^5)$. Consistency: at $x_t = \mu_x$ the forecast is the mean, e.g. case i: $5.2156 + 10(.73872 - .26028) = 10 \checkmark$. Case v's formula is valid although the process is not stationary.

(d) By (2.4.18), $E_t \sum_k .95^k y_{t+k} = G(I - .95A)^{-1} x_t$. Because $\alpha \neq 0$ the right side also carries a **constant** $\tilde{h}_c$ (the last element), which the printed form omits:

case	$\tilde{h}_c$	$\tilde{h}_0$	$\tilde{h}_1$	$\tilde{h}_2$	$\tilde{h}_3$
i, ii	145.3155	7.64818	-2.17973	0	0
iii	65.5172	6.89655	0	0	0
iv	70.7641	2.48295	1.06441	1.12043	1.17940
v	undefined: $.95 \times 1.0782 = 1.0243 > 1$, the sum diverges				

Case iii: $\tilde{h}_0 = \frac{1}{1-.95(.9)} = \frac{1}{.145} = 6.89655$. Consistency at the mean: case i, $145.3155 + 10(7.64818 - 2.17973) = 200 = \frac{10}{1-.95} \checkmark$; case iv, $70.7641 + 5(5.84719) = 100 \checkmark$.

(e) $E(y_t - \mu_y)(y_{t-k} - \mu_y) = GA^k \Sigma_\infty G'$ (2.4.12):

case	$k = 1$	$k = 5$	$k = 10$
i	6.857143	3.702857	1.597579
ii	27.428571	14.811429	6.390317
iii	4.736842	3.107842	1.835150
iv	.415887	.365232	.131579
v	not defined		

Case iii is $.9^k \times 5.263158$. Case iv has a larger autocovariance at lag 5 than one might expect from $\rho_1 = .2$ alone: the $\rho_4 = .5$ term is a “seasonal” at lag 4.

Verification. `ls_ch02.ipynb`, §2.4: Σ_∞ by doubling against $\sum_j (GA^j C)^2$; all autocovariances against $\sum_j \psi_j \psi_{j+k}$; $(I - .95A)^{-1}$ against the truncated sum; the AR(2) and AR(1) closed forms. $\checkmark$

Exercise 2.5 Value of a Consumption Process

Question

A consumer's rate of consumption follows the stochastic process

$$\begin{aligned} c_{t+1} &= \alpha_c + \sum_{j=1}^2 \rho_j c_{t-j+1} + \sum_{j=1}^2 \delta_j z_{t+1-j} + \psi_1 w_{1,t+1} \\ z_{t+1} &= \sum_{j=1}^2 \gamma_j c_{t-j+1} + \sum_{j=1}^2 \phi_j z_{t-j+1} + \psi_2 w_{2,t+1} \end{aligned} \quad (1)$$

where w_{t+1} is a 2×1 martingale difference sequence, adapted to $J_t = [w_t \dots w_1 c_0 c_{-1} z_0 z_{-1}]$, with contemporaneous covariance matrix $E w_{t+1} w'_{t+1} | J_t = I$,

and the coefficients $\rho_j, \delta_j, \gamma_j, \phi_j$ are such that the matrix $A = \begin{bmatrix} \rho_1 & \rho_2 & \delta_1 & \delta_2 & \alpha_c \\ 1 & 0 & 0 & 0 & 0 \\ \gamma_1 & \gamma_2 & \phi_1 & \phi_2 & 0 \\ 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 1 \end{bmatrix}$

has eigenvalues bounded strictly below unity in modulus. The consumer evaluates consumption streams according to $V_0 = E_0 \sum_{t=0}^{\infty} .95^t u(c_t)$ (2), where the one-period utility function is $u(c_t) = -.5(c_t - 60)^2$ (3).

- (a) Find a formula for V_0 in terms of the parameters of the one-period utility function (3) and the stochastic process for consumption.
- (b) Compute V_0 for the following two sets of parameter values: i. $\rho = [.8 \ - .3]$, $\alpha_c = 1$, $\delta = [.2 \ 0]$, $\gamma = [0 \ 0]$, $\phi = [.7 \ -.2]$, $\psi_1 = \psi_2 = 1$. ii. Same as for part i except now $\psi_1 = 2, \psi_2 = 1$. *Hint:* Remember `doublej.m`.

Solution:

- (a) State $x_t = [c_t, c_{t-1}, z_t, z_{t-1}, 1]'$, $x_{t+1} = A x_t + C w_{t+1}$ with $C = \begin{bmatrix} \psi_1 & 0 & 0 & 0 & 0 \\ 0 & 0 & \psi_2 & 0 & 0 \end{bmatrix}'$. Write $c_t - 60 = H x_t$ with $H = [1 \ 0 \ 0 \ 0 \ -60]$, so $u(c_t) = -.5 x_t' Y x_t$ with $Y = H' H$. With $\beta = .95$,

$$V_0 = -.5 E_0 \sum_{t=0}^{\infty} \beta^t x_t' Y x_t = -.5 (x_0' \nu x_0 + \sigma),$$

where, by (2.4.19)–(2.4.20),

$$\boxed{\nu = H' H + \beta A' \nu A, \quad \sigma = \frac{\beta}{1 - \beta} \text{trace}(\nu C C').}$$

ν solves a discrete Lyapunov equation in $\sqrt{\beta} A'$ (solvable: the eigenvalues of $\sqrt{\beta} A$ are at most $\sqrt{.95} < 1$ in modulus, the constant included). An equivalent mean–variance form, from $u = -.5[(E c_t - 60)^2 + \text{Var} c_t]$ in expectation, is $V_0 = -.5 \sum_t \beta^t [(H \mu_t)^2 + H \Sigma_t H']$ with $\mu_{t+1} = A \mu_t$, $\Sigma_{t+1} = A \Sigma_t A' + C C'$, $\mu_0 = x_0$, $\Sigma_0 = 0$.

- (b) The exercise does not fix x_0 . We evaluate at the stationary mean, $x_0 = \mu_x$: from row 1 of $(I - A)\mu = 0$ with $\mu_z = 0$ (the z equation has no constant and $\gamma = 0$), $\mu_c = 1 + .8\mu_c - .3\mu_c \Rightarrow \mu_c = \frac{1}{.5} = 2$, so $\mu_x = [2, 2, 0, 0, 1]'$.

case	Var(c) stationary	σ	$V_0(x_0 = \mu_x)$	$E V_0, x_0 \sim \text{stationary}$
i ($\psi_1 = 1$)	1.970067	36.1441	$\boxed{-33658.07}$	-33659.70
ii ($\psi_1 = 2$)	7.276189	134.3401	$\boxed{-33707.17}$	-33712.76

Decomposition: at $x_0 = \mu_x$, $-.5 x_0' \nu x_0 = -.5 \frac{(2-60)^2}{1-.95} = -.5(3364)(20) = -33640$; the remainder $-.5\sigma$ (-18.07 and -67.17) is the cost of future volatility. Check of the last column: with x_0 stationary, $V = -.5 \frac{(\mu_c - 60)^2 + \text{Var}c}{1-\beta}$, e.g. case i $-.5(3364 + 1.970067)(20) = -33659.70 \checkmark$.

Reading. Almost all the loss comes from mean consumption being far below the bliss point 60; doubling ψ_1 roughly quadruples the volatility cost.

Verification. `ls_ch02.ipynb`, §2.5: ν by doubling against the deterministic mean-variance sum over 2500 periods. $\checkmark$

Exercise 2.6 Stationary Moments and a Population Regression

Question

Consider the stochastic process $\{c_t, z_t\}$ defined by equations (1) in exercise 2.5. Assume the parameter values described in part b, item i. If possible, assume the initial conditions are such that $\{c_t, z_t\}$ is covariance stationary.

- (a) Compute the initial mean and covariance matrix that make the process covariance stationary.
- (b) For the initial conditions in part a, compute numerical values of the following population linear regression: $c_{t+2} = \alpha_0 + \alpha_1 z_t + \alpha_2 z_{t-4} + w_t$ where $E w_t [1 \ z_t \ z_{t-4}] = [0 \ 0 \ 0]$.

Solution:

- (a) Stationarity is possible: the AR blocks have roots of $\lambda^2 - .8\lambda + .3$ (modulus $\sqrt{.3} = .548$) and $\lambda^2 - .7\lambda + .2$ (modulus $\sqrt{.2} = .447$). Mean (Exercise 2.5): $\mu_x = [2, 2, 0, 0, 1]'$. Covariance: $\Sigma_\infty = A\Sigma_\infty A' + CC'$; for $[c_t, c_{t-1}, z_t, z_{t-1}]$,

$$\Sigma_\infty = \begin{bmatrix} 1.970067 & 1.249352 & .240522 & .486912 \\ 1.249352 & 1.970067 & .070983 & .240522 \\ .240522 & .070983 & 1.578947 & .921053 \\ .486912 & .240522 & .921053 & 1.578947 \end{bmatrix}$$

(zero row and column for the constant). Checks on the z block, an AR(2) that c does not feed ($\gamma = 0$): $\text{Var}z = \frac{1.2}{.8(1.44-.49)} = \frac{1.2}{.76} = 1.578947$ and $\text{Cov}(z_t, z_{t-1}) = \frac{\phi_1}{1-\phi_2} \text{Var}z = \frac{.7}{1.2}(1.578947) = .921053 \checkmark$.

- (b) Since $\mu_z = 0$, $\alpha_0 = \mu_c - (\alpha_1 + \alpha_2)\mu_z = \boxed{2}$, and (α_1, α_2) solve the normal equations

built from $\text{Cov}(x_{t+j}, x_t) = A^j \Sigma_\infty$:

$$\begin{bmatrix} \gamma_z(0) & \gamma_z(4) \\ \gamma_z(4) & \gamma_z(0) \end{bmatrix} \begin{bmatrix} \alpha_1 \\ \alpha_2 \end{bmatrix} = \begin{bmatrix} \text{Cov}(c_{t+2}, z_t) \\ \text{Cov}(c_{t+2}, z_{t-4}) \end{bmatrix},$$

$$\begin{bmatrix} 1.578947 & -.033553 \\ -.033553 & 1.578947 \end{bmatrix} \begin{bmatrix} \alpha_1 \\ \alpha_2 \end{bmatrix} = \begin{bmatrix} .501584 \\ -.049666 \end{bmatrix}.$$

Solving: $\alpha_1 = .317144, \alpha_2 = -.024716$.

Reading. z Granger-causes c through $\delta_1 = .2$; z_{t-4} gets a small negative weight because, given z_t , it corrects for the hump-shaped propagation.

Verification. `ls_ch02.ipynb`, §2.6: OLS on a 400,000-period simulation gives (1.9967, .3200, -.0247). ✓

Exercise 2.8 Cagan Money Demand under Perfect Foresight

Question

This exercise deals with Cagan's money demand under rational expectations. A version of Cagan's (1956) demand function for money is $m_t - p_t = -\alpha(p_{t+1} - p_t)$, $\alpha > 0$, $t \geq 0$ (1), where m_t is the log of the nominal money supply and p_t is the price level at t . Equation (1) states that the demand for real balances varies inversely with the expected rate of inflation, $(p_{t+1} - p_t)$. There is no uncertainty, so the expected inflation rate equals the actual one. The money supply obeys the difference equation $(1 - L)(1 - \rho L)m_t^s = 0$ (2) subject to initial condition for m_{-1}^s, m_{-2}^s . In equilibrium, $m_t \equiv m_t^s \forall t \geq 0$ (3) (i.e., the demand for money equals the supply). For now assume that $|\rho\alpha/(1 + \alpha)| < 1$ (4). An *equilibrium* is a $\{p_t\}_{t=0}^\infty$ that satisfies equations (1), (2), and (3) for all t .

- Find an expression for an equilibrium p_t of the form $p_t = \sum_{j=0}^n w_j m_{t-j} + f_t$ (5). Please tell how to get formulas for the w_j for all j and the f_t for all t .
- How many equilibria are there?
- Is there an equilibrium with $f_t = 0$ for all t ?
- Briefly tell where, if anywhere, condition (4) plays a role in your answer to part a.
- For the parameter values $\alpha = 1, \rho = 1$, compute and display all the equilibria.

Solution:

- Step 1: solve (1) forward.* Rearranging (1), $(1 + \alpha)p_t = m_t + \alpha p_{t+1}$, i.e.

$$p_t = (1 - \lambda)m_t + \lambda p_{t+1}, \quad \lambda \equiv \frac{\alpha}{1 + \alpha} \in (0, 1).$$

Iterating T times, $p_t = (1 - \lambda) \sum_{j=0}^{T-1} \lambda^j m_{t+j} + \lambda^T p_{t+T}$. Every solution is a particular

solution plus a solution of the homogeneous equation $f_t = \lambda f_{t+1}$:

$$p_t = (1 - \lambda) \sum_{j=0}^{\infty} \lambda^j m_{t+j} + f_t, \quad f_t = c \lambda^{-t} = c \left(\frac{1 + \alpha}{\alpha} \right)^t, \quad c \in \mathbb{R}.$$

Step 2: forecast money. From (2), $\Delta m_t = \rho \Delta m_{t-1}$, so $\Delta m_{t+k} = \rho^k \Delta m_t$ and $m_{t+j} = m_t + \Delta m_t \sum_{k=1}^j \rho^k$.

Step 3: sum. Exchanging the order of summation,

$$\begin{aligned} \sum_{j \geq 0} \lambda^j m_{t+j} &= \frac{m_t}{1 - \lambda} + \Delta m_t \sum_{k \geq 1} \rho^k \sum_{j \geq k} \lambda^j = \frac{m_t}{1 - \lambda} + \Delta m_t \sum_{k \geq 1} \frac{(\lambda \rho)^k}{1 - \lambda} \\ &= \frac{m_t}{1 - \lambda} + \frac{\Delta m_t}{1 - \lambda} \frac{\lambda \rho}{1 - \lambda \rho}. \end{aligned}$$

Step 4: collect. Multiplying by $1 - \lambda$ and using $\frac{\lambda \rho}{1 - \lambda \rho} = \frac{\alpha \rho / (1 + \alpha)}{(1 + \alpha - \alpha \rho) / (1 + \alpha)} = \frac{\alpha \rho}{1 + \alpha - \alpha \rho}$:

$$\boxed{\begin{aligned} p_t &= w_0 m_t + w_1 m_{t-1} + f_t, \quad f_t = c \left(\frac{1 + \alpha}{\alpha} \right)^t, \\ w_0 &= 1 + \frac{\alpha \rho}{1 + \alpha - \alpha \rho}, \quad w_1 = -\frac{\alpha \rho}{1 + \alpha - \alpha \rho}, \quad w_j = 0 \quad (j \geq 2) \end{aligned}}$$

so $n = 1$. General recipe: $w(L)$ comes from applying the forward operator $(1 - \lambda)(1 - \lambda L^{-1})^{-1}$ to the money process and using (2) to replace future m 's by m_t, m_{t-1} (the Hansen–Sargent formula); f_t solves the homogeneous forward equation.

- (b) A continuum: one equilibrium for each $c \in \mathbb{R}$ (each value of p_0). The definition imposes no terminal condition, so the explosive “bubble” paths $c \lambda^{-t}$ are not ruled out.
- (c) Yes: $c = 0$, the *fundamental* (bubble-free) equilibrium, in which p_t is the present value of current and future money supplies.
- (d) Condition (4), $|\lambda \rho| < 1$, is exactly what makes the geometric series in Step 3 converge (and justifies exchanging sums): it bounds the growth of expected money relative to the discount λ . Without it the fundamental solution does not exist and (5) is meaningless. It is not needed for the bubble component.
- (e) $\alpha = \rho = 1$: $\lambda = \frac{1}{2}$, $\lambda \rho = \frac{1}{2}$ satisfies (4); $w_0 = 1 + \frac{1}{2-1} = 2$, $w_1 = -1$. With $\rho = 1$ money grows linearly: $m_t = m_{-1} + (t + 1)g$, $g \equiv m_{-1} - m_{-2}$. Hence

$$\boxed{p_t = 2m_t - m_{t-1} + c 2^t = m_t + g + c 2^t, \quad c \in \mathbb{R}.}$$

Check in (1): $m_t - p_t = -g - c 2^t$ and $-(p_{t+1} - p_t) = -(g + c 2^t) \checkmark$. In the fundamental equilibrium the price level runs one period of money growth ahead of money (inflation = g , real balances $-g$); every $c \neq 0$ adds an explosive bubble that doubles each period. Figure 1 displays the equilibria for $m_{-2} = 0$, $m_{-1} = 1$.

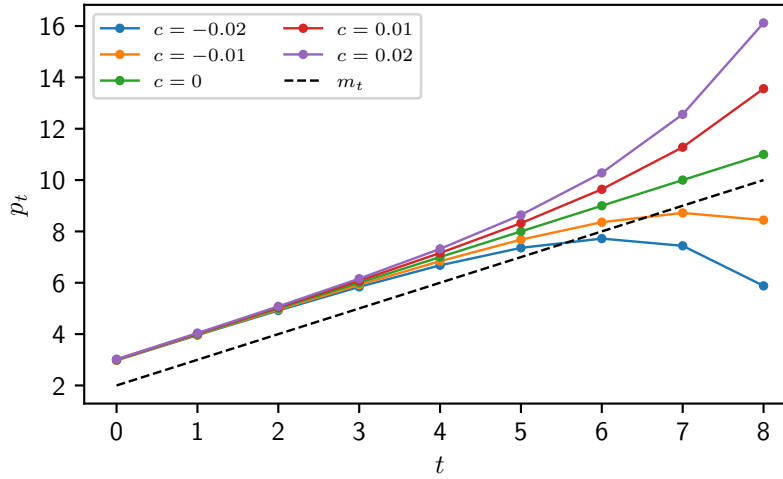


Figure 1: Exercise 2.8(e): equilibrium price paths $p_t = m_t + g + c2^t$ for five values of c ($\alpha = \rho = 1$, $m_{-2} = 0$, $m_{-1} = 1$).

Verification. `ls_ch02.ipynb`, § 2.8: (w_0, w_1) against the brute-force forward sum truncated at 700 terms for $(\alpha, \rho) \in \{(2, .6), (1, 1), (3, .9)\}$; (1) holds on every plotted path. ✓

Exercise 2.9 Pricing with a Stochastic Discount Factor

Question

The $n \times 1$ state vector of an economy is governed by the linear stochastic difference equation $x_{t+1} = Ax_t + C_t w_{t+1}$ (1) where C_t is a possibly time-varying matrix (known at t) and w_{t+1} is an $m \times 1$ martingale difference sequence adapted to its own history with $E w_{t+1} w'_{t+1} | J_t = I$, where $J_t = [w_t \dots w_1 x_0]$. A scalar one-period payoff p_{t+1} is given by $p_{t+1} = P x_{t+1}$ (2). The stochastic discount factor for this economy is a scalar m_{t+1} that obeys $m_{t+1} = \frac{M x_{t+1}}{M x_t}$ (3). Finally, the price at time t of the one-period payoff is given by $q_t = f_t(x_t)$, where f_t is some possibly time-varying function of the state. That m_{t+1} is a stochastic discount factor means that $E(m_{t+1} p_{t+1} | J_t) = q_t$ (4).

- Compute $f_t(x_t)$, describing in detail how it depends on A and C_t .
- Suppose that an econometrician has a time series data set $X_t = [z_t \ m_{t+1} \ p_{t+1} \ q_t]$, for $t = 1, \dots, T$, where z_t is a strict subset of the variables in the state x_t . Assume that investors in the economy see x_t even though the econometrician sees only a subset z_t of x_t . Briefly describe a way to use these data to test implication (4). (Possibly but perhaps not useful hint: recall the law of iterated expectations.)

Solution:

- M and P are $1 \times n$, so $M x_{t+1}$ and $P x_{t+1}$ are scalars and $(M x_{t+1})(P x_{t+1}) =$

$Mx_{t+1}x'_{t+1}P'$. Since Mx_t is known at t ,

$$q_t = \frac{M E[x_{t+1}x'_{t+1}|J_t] P'}{Mx_t}.$$

From (1), $E[x_{t+1}x'_{t+1}|J_t] = Ax_t x'_t A' + C_t E[w_{t+1}w'_{t+1}|J_t] C'_t = Ax_t x'_t A' + C_t C'_t$ (the cross terms vanish because $E[w_{t+1}|J_t] = 0$). Therefore

$$\boxed{f_t(x_t) = \frac{(MAx_t)(PAx_t) + MC_t C'_t P'}{Mx_t}} = \underbrace{E_t[m_{t+1}] E_t[p_{t+1}]}_{\text{"risk-neutral" part}} + \underbrace{\text{Cov}_t(m_{t+1}, p_{t+1})}_{\text{risk adjustment}}.$$

A enters through the conditional means MAx_t, PAx_t ; C_t only through the conditional covariance $MC_t C'_t P'/(Mx_t)$, which is the **only** source of time variation in f_t beyond x_t itself. If $C_t = 0$ the price is the discounted certainty equivalent. (We need $Mx_t \neq 0$.)

- (b) Define the pricing error $e_{t+1} = m_{t+1}p_{t+1} - q_t$; (4) says $E[e_{t+1}|J_t] = 0$. The econometrician's information $\mathcal{I}_t = \{z_t, X_{t-1}, X_{t-2}, \dots\}$ is contained in J_t , so by the law of iterated expectations $E[e_{t+1}|\mathcal{I}_t] = E\{E[e_{t+1}|J_t]|\mathcal{I}_t\} = 0$. Hence for any vector of instruments h_t measurable with respect to $\mathcal{I}_t$ (e.g. $h_t = [1, z_t, m_t, p_t, q_{t-1}]'$), $E[e_{t+1}h_t] = 0$. All pieces of e_{t+1} are in the data, even though x_t is not. Form $g_T = \frac{1}{T} \sum_t e_{t+1}h_t$ and test $g_T \approx 0$ with Hansen's GMM J -statistic $Tg'_T \hat{S}^{-1} g_T \sim \chi^2(\dim h)$, $\hat{S}$ a HAC estimate of the long-run variance of $e_{t+1}h_t$ (by (4), $e_{t+1}h_t$ is itself a martingale difference, so no lags are needed).

Verification. `ls_ch02.ipynb`, §2.9: formula -13.5407 vs. Monte Carlo (2×10^6 draws) -13.5340 at a random state. ✓

Exercise 2.29 Pure Prediction

Question

Consider the Bellman equation $\mu_t = Rx_t + A'\mu_{t+1}$ (1), where $x_{t+1} = Ax_t$. Here x_t is an $n \times 1$ vector, A is an $n \times n$ stable matrix, and R is an $n \times n$ positive definite matrix.

- Show that $\mu_t = R \sum_{j=0}^{\infty} A'^j x_{t+j}$ (2) is a solution of the forward-looking difference equation (1).
- Guess that $\mu_t = Px_t$, then verify that $P = R + A'PA$ (3).
- Please compare formula (3) for P with formula (2.4.10) for the unconditional covariance matrix of a vector governed by an autoregression.
- Please compare formula (3) with the formula (2.4.25) for P that is a key step in the Howard policy improvement algorithm.
- Can you invent a counterpart of the Howard policy improvement algorithm to compute a time-invariant version of the Kalman filter?

Solution:

- (a) The correct solution has R *inside* the sum: $\mu_t = \sum_{j=0}^{\infty} A^j R x_{t+j}$. Check:

$$R x_t + A' \mu_{t+1} = R x_t + \sum_{j \geq 0} A^{j+1} R x_{t+1+j} = R x_t + \sum_{k \geq 1} A^k R x_{t+k} = \mu_t. \checkmark$$

The sum converges because $x_{t+k} = A^k x_t$ and $\|A^k R A^k\| \leq c r^{2k}$ for any r above the spectral radius of A , which is < 1 . The printed form $R \sum_j A^j x_{t+j}$ gives $R x_t + A' R \sum_j A^j x_{t+1+j}$, which equals it only if $A' R = R A'$; see the trap box.

- (a') Substituting $x_{t+j} = A^j x_t$: $\mu_t = \left(\sum_{j \geq 0} A^j R A^j \right) x_t \equiv P x_t$. Then $P - A' P A = \sum_{j \geq 0} A^j R A^j - \sum_{j \geq 1} A^j R A^j = R$, i.e. $\boxed{P = R + A' P A}$. Directly from (1): $P x_t = R x_t + A' P A x_t$ for all x_t gives the same equation.
- (c) (2.4.10) is $\Sigma_{\infty} = A \Sigma_{\infty} A' + C C'$, with solution $\sum_j A^j C C' A^j$. It is the same discrete Lyapunov equation with $A \leftrightarrow A'$ and $C C' \leftrightarrow R$: the **dual**. Σ_{∞} accumulates past shocks forward through A ; P accumulates future payoffs backward through A' .
- (d) (2.4.25) is $P_0 = (R + F_0' Q F_0) + \left(\sqrt{\beta} (A - B F_0) \right)' P_0 \left(\sqrt{\beta} (A - B F_0) \right)$: formula (3) with $R \rightarrow R + F_0' Q F_0$ and $A \rightarrow \sqrt{\beta} (A - B F_0)$. The policy-evaluation step of Howard's algorithm is a pure prediction problem: the value of a fixed rule is a discounted sum of future quadratic payoffs along the closed-loop law of motion.
- (e) Yes, by duality. *Evaluation*: for a fixed gain K with $A - K G$ stable, the steady-state error $e_t = x_t - \hat{x}_t$ obeys $e_{t+1} = (A - K G) e_t + C w_{t+1} - K v_t$, so its covariance solves the Lyapunov equation

$$\Sigma_K = (A - K G) \Sigma_K (A - K G)' + C C' + K R K' \quad (\text{cf. (2.7.14d) with } K \text{ held fixed}).$$

Improvement: given Σ_K , choose the gain that minimises next period's error covariance $(A - K G) \Sigma_K (A - K G)' + K R K'$; the first-order condition $-2 A \Sigma_K G' + 2 K (G \Sigma_K G' + R) = 0$ gives $K_{\text{new}} = A \Sigma_K G' (G \Sigma_K G' + R)^{-1}$ (2.7.14b). Iterate until K settles, starting from a K_0 with $A - K_0 G$ stable ($K_0 = 0$ if A is stable). This mirrors (2.4.30) under the duality $A \rightarrow A'$, $B \rightarrow G'$, $R \rightarrow C C'$, $Q \rightarrow R$, $F \rightarrow K'$, $\beta = 1$.

Verification. `ls_ch02.ipynb`, § 2.29: P from a Lyapunov solver against $\sum_{j < 400} A^j R A^j$; residual of (1) is 4×10^{-14} with the $A^j R$ order and 9.4 with the printed order; the Howard-Kalman iteration converges in 6 steps to the Riccati fixed point for two systems. $\checkmark$

Where this goes wrong

Formula (2) as printed, $\mu_t = R \sum_j A^j x_{t+j}$, is not a solution of (1) unless R commutes with A' (e.g. $R = rI$ or scalar x). The weighting matrix must sit between the powers of A' and the future state: $\mu_t = \sum_j A^j R x_{t+j}$. The notebook shows the printed version

failing (1) for a random 3×3 example.

Spectra

Exercise 2.7 Simulations, Impulse Responses and Spectra

Question

Get the Matlab programs `bigshow3.m` and `freq.m` from www.tomsargent.com/source_code/mitbook.zip. Use `bigshow3` to compute and display a simulation of length 80, an impulse response function, and a spectrum for each of the following scalar stochastic processes y_t . In each of the following, w_t is a scalar martingale difference sequence adapted to its own history and the initial values of lagged y 's.

- a. $y_t = w_t$. b. $y_t = (1 + .5L)w_t$. c. $y_t = (1 + .5L + .4L^2)w_t$. d. $(1 - .999L)y_t = (1 - .4L)w_t$. e. $(1 - .8L)y_t = (1 + .5L + .4L^2)w_t$. f. $(1 + .8L)y_t = w_t$. g. $y_t = (1 - .6L)w_t$.

Study the output and look for patterns. When you are done, you will be well on your way to knowing how to read spectral densities.

Solution:

Write each process as $a(L)y_t = b(L)w_t$ with $EW_t^2 = 1$. The impulse response is $h(L) = b(L)/a(L)$, and by (2.11.2) with $y = Gx$ the spectrum is

$$S_y(\omega) = h(e^{-i\omega})h(e^{i\omega}) = \frac{|b(e^{-i\omega})|^2}{|a(e^{-i\omega})|^2}, \quad S_y(0) = \frac{b(1)^2}{a(1)^2}, \quad S_y(\pi) = \frac{b(-1)^2}{a(-1)^2}.$$

The Python counterpart of `bigshow3` (`ls_ch02.ipynb`, § 2.7) produces Figure 2.

	$\text{Var}(y) = \sum h_j^2$	$S(0)$	$S(\pi)$	shape
a	1	1	1	flat
b	1.25	2.25	.25	falls from 0 to π
c	1.41	3.61	.81	falls, dip near $\omega \approx 2$
d	180.49	360000	.49	enormous spike at $\omega = 0$
e	8.45	90.25	.25	low-frequency power, dip near $\omega \approx 2.1$
f	2.78	.31	.25	rises to a peak at π
g	1.36	.16	2.56	rises from 0 to π

Patterns.

- *White noise (a)* has a flat spectrum: every frequency contributes equally, the impulse response is a single spike, and the path has no visible persistence.
- *Positive MA coefficients (b, c)* smooth the noise: neighbouring y 's share shocks, the impulse response is cut off after q lags, and power moves to low frequencies. A zero of $b(z)$ near the unit circle creates a trough in S_y ; in (c) the zeros of $1 + .5z + .4z^2$ have modulus $1/\sqrt{.4} = 1.58$ and angle ≈ 2 , hence the dip there.

- *Near unit root (d)*. $(1 - .999L)^{-1}$ makes the impulse response almost flat at $.999 - .4 = .599$ after lag 0, and $S_y(0) = (.6/.001)^2 = 360000$: nearly all variance sits at very low frequency. This is Granger's "typical spectral shape" in its extreme form; the simulation wanders. It is the Muth process of Figure 2.11.4 and section 2.10.1, where most of a shock is permanent.
- *Mixed (e)*. The AR(1) factor adds low-frequency power and the MA factor keeps the dip of (c); the impulse response is hump-shaped (peaks at lag 2).
- *Negative AR root (f)*. $(1 + .8L)^{-1}$ gives an impulse response alternating in sign, $(-.8)^j$, so the path zig-zags and power piles up at $\omega = \pi$ (period 2).
- *Negative MA coefficient (g)*. Opposite of (b): it is a partial difference, so it removes low-frequency power, $S(0) = (1 - .6)^2 = .16$.

Rule of thumb: smooth, persistent paths and slowly decaying same-sign impulse responses mean power at low frequencies; choppy paths and sign-alternating responses mean power near π ; a spectral peak inside $(0, \pi)$ means a pseudo-cycle (Exercise 2.15).

Verification. `ls_ch02.ipynb`, § 2.7: for each process, $\frac{1}{2\pi} \int_{-\pi}^{\pi} S_y$ equals $\sum_j h_j^2$ to relative error 10^{-4} . ✓

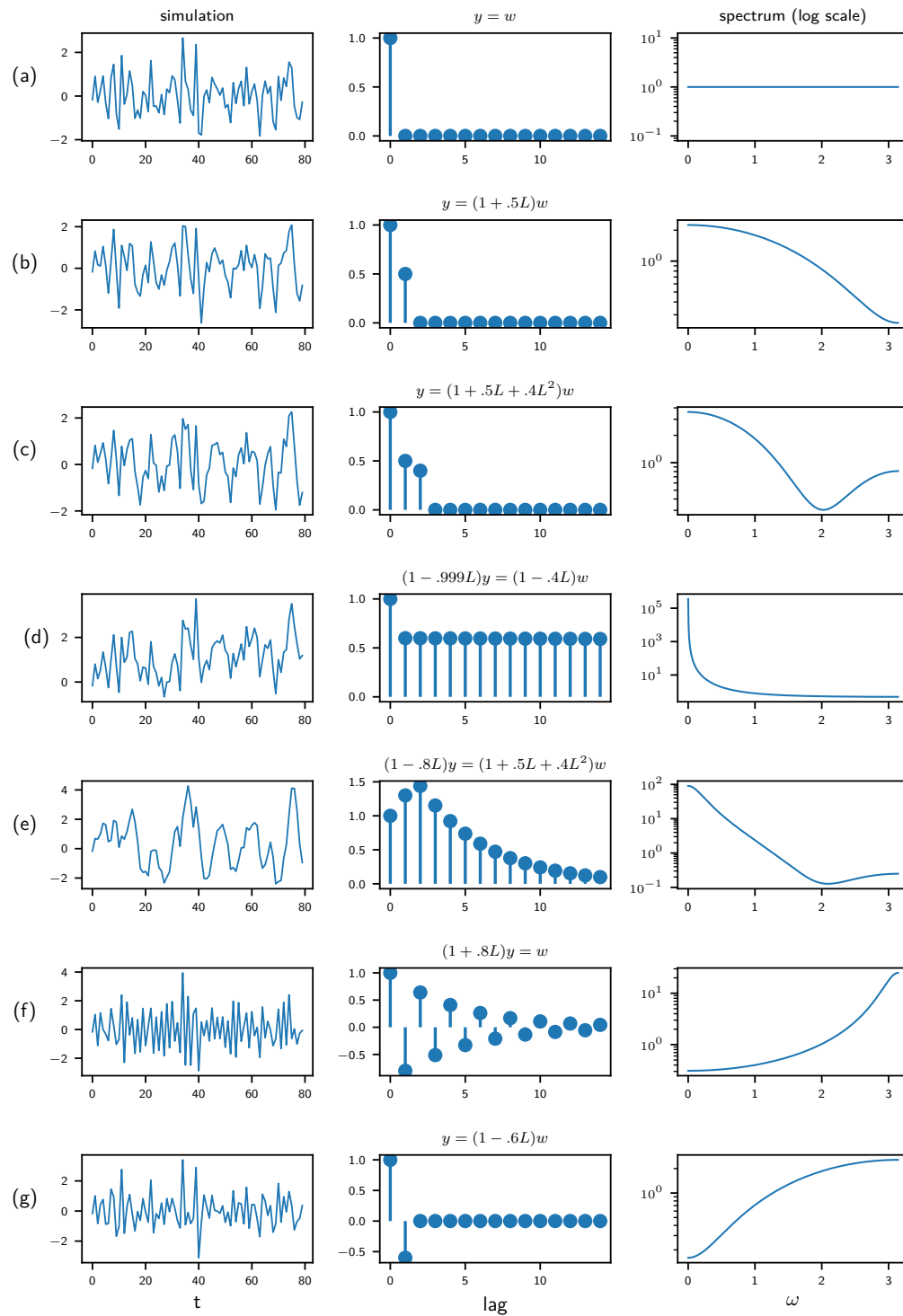


Figure 2: Exercise 2.7: simulation (length 80, common shocks), impulse response and spectrum (log scale) for processes a–g.

Exercise 2.15 The Slutsky Effect

Question

Suppose that a scalar is related to a scalar white noise w_t with variance 1 by $y_t = h(L)w_t$ where $h(L) = \sum_{j=0}^{\infty} L^j h_j$ and $\sum_{j=0}^{\infty} h_j^2 < +\infty$. Then a special case of formula (2.11.2) coupled with the observer equation $y_t = Gx_t$ implies that the spectrum of y is given by $S_y(\omega) = h(\exp(-i\omega))h(\exp(i\omega)) = |h(\exp(-i\omega))|^2$ where $h(\exp(-i\omega)) = \sum_{j=0}^{\infty} h_j \exp(-i\omega j)$. In a famous paper, Slutsky investigated the consequences of applying the following filter to white noise: $h(L) = (1 + L)^n(1 - L)^m$ (i.e., the convolution of n two-period moving averages with m difference operators). Compute and plot the spectrum of y for $\omega \in [-\pi, \pi]$ for the following choices of m, n :

(a) $m = 10, n = 10$. (b) $m = 10, n = 40$. (c) $m = 40, n = 10$. (d) $m = 120, n = 30$.

(e) Comment on these results. *Hint:* Notice that $h(\exp(-i\omega)) = (1 + \exp(-i\omega))^n(1 - \exp(-i\omega))^m$.

Solution:

Closed form. $|1 + e^{-i\omega}|^2 = (1 + \cos \omega)^2 + \sin^2 \omega = 2 + 2 \cos \omega = 4 \cos^2 \frac{\omega}{2}$ and $|1 - e^{-i\omega}|^2 = 2 - 2 \cos \omega = 4 \sin^2 \frac{\omega}{2}$. Hence

$$S_y(\omega) = (2 + 2 \cos \omega)^n (2 - 2 \cos \omega)^m = 4^{n+m} \cos^{2n} \frac{\omega}{2} \sin^{2m} \frac{\omega}{2},$$

symmetric in ω , with $S_y(0) = 0$ (for $m \geq 1$) and $S_y(\pm\pi) = 0$ (for $n \geq 1$).

Peak. $\frac{d}{d\omega} \log S_y = -n \tan \frac{\omega}{2} + m \cot \frac{\omega}{2} = 0 \Rightarrow \tan^2 \frac{\omega}{2} = \frac{m}{n}$, so

$$\boxed{\omega^* = 2 \arctan \sqrt{m/n}}.$$

Sharpness. $\frac{d^2}{d\omega^2} \log S_y = -\frac{n}{2} \sec^2 \frac{\omega}{2} - \frac{m}{2} \csc^2 \frac{\omega}{2}$; at ω^* , $\sec^2 = \frac{n+m}{n}$ and $\csc^2 = \frac{n+m}{m}$, so the curvature is $-(n+m)$ and $S_y \approx S_y(\omega^*) e^{-(n+m)(\omega - \omega^*)^2/2}$: a Gaussian bump of width $1/\sqrt{n+m}$, with half-power width $2\sqrt{2 \ln 2/(n+m)}$.

	m	n	ω^*	period	half-power width (num./formula)	$\log_{10} S(\omega^*)$
a	10	10	$\pi/2 = 1.5708$	4.00	.5235 / .5265	6.02
b	10	40	.9273	6.78	.3319 / .3330	19.24
c	40	10	2.2143	2.84	.3319 / .3330	19.24
d	120	30	2.2143	2.84	.1920 / .1923	57.71

(e) *Comment.* Summing (n) is a low-pass filter; differencing (m) is high-pass. Composing them gives a band-pass filter whose output has a single sharp spectral peak inside $(0, \pi)$, i.e. a nearly periodic “cycle” of period $2\pi/\omega^*$, although the input is serially uncorrelated. This is the **Slutsky effect**: mechanical smoothing and differencing of pure noise can manufacture regular-looking business cycles. The ratio m/n locates the cycle (more differencing, a shorter period: (b) vs. (c)); the total $m+n$ sets how regular it is (same ω^* in (c) and (d), but (d) is much narrower). The levels are astronomically large (10^{57} in (d)), so the plot normalises each spectrum by its maximum (Figure 3).

Verification. 1s_ch02.ipynb, § 2.15: numerical argmax on a grid of 200,001 points matches ω^* within 10^{-4} . ✓

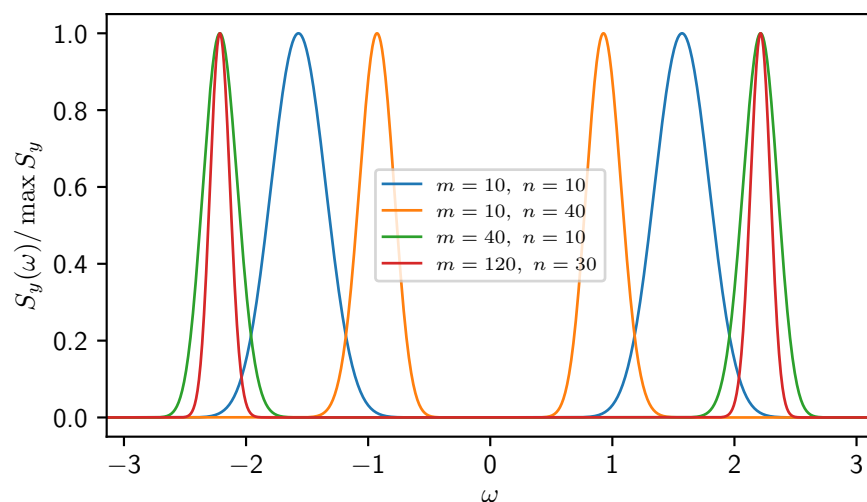


Figure 3: Exercise 2.15: normalised spectra $S_y(\omega)/\max S_y$ of Slutsky's filter.

The Kalman Filter, Innovations and VARs

Exercise 2.19 IQ

Question

An infinitely lived person's 'true intelligence' $\theta \sim \mathcal{N}(100, 100)$, i.e., mean 100, variance 100. For each date $t \geq 0$, the person takes a 'test' with the outcome being a univariate random variable $y_t = \theta + v_t$, where v_t is an iid process with distribution $\mathcal{N}(0, 100)$. The person's initial IQ is $IQ_0 = 100$ and at date $t \geq 1$ before the date t test is taken it is $IQ_t = E\theta|y^{t-1}$, where y^{t-1} is the history of test scores from date 0 until date $t-1$.

- Give a recursive formula for IQ_t and for $E(IQ_t - \theta)^2$.
- Use Matlab to simulate 10 draws of θ and associated paths of y_t, IQ_t for $t = 0, \dots, 50$.
- Prove that $\lim_{t \rightarrow \infty} E(IQ_t - \theta)^2 = 0$.

Solution:

This is the Kalman filter (2.7.14) with $x_t = \theta$, $A = 1$, $C = 0$, $G = 1$, $R = 100$, $\hat{x}_0 = 100$, $\Sigma_0 = 100$, and $\hat{x}_t = IQ_t$ (Jovanovic's application, section 2.10.2).

(a) From (2.7.14b,c,d) with $A = 1, C = 0$:

$$K_t = \frac{\Sigma_t}{\Sigma_t + 100}, \quad \boxed{\text{IQ}_{t+1} = \text{IQ}_t + K_t(y_t - \text{IQ}_t)},$$

$$\boxed{\Sigma_{t+1} = (1 - K_t)^2 \Sigma_t + K_t^2 100 = \frac{100 \Sigma_t}{\Sigma_t + 100}},$$

with $\Sigma_t = E(\text{IQ}_t - \theta)^2$. *Closed forms.* Inverting, $\frac{1}{\Sigma_{t+1}} = \frac{1}{\Sigma_t} + \frac{1}{100}$ (2.10.9), so $\frac{1}{\Sigma_t} = \frac{t+1}{100}$, i.e. $\Sigma_t = \frac{100}{t+1}$ and $K_t = \frac{1}{t+2}$. Then $(t+2)\text{IQ}_{t+1} = (t+1)\text{IQ}_t + y_t$; telescoping from $\text{IQ}_0 = 100$:

$$\text{IQ}_t = \frac{100 + \sum_{s=0}^{t-1} y_s}{t+1}.$$

Because the prior variance equals the noise variance, the prior counts as exactly one extra test with score 100.

(b) Figure 4: ten draws of θ and the implied IQ paths.

(c) $E(\text{IQ}_t - \theta)^2 = \Sigma_t = \frac{100}{t+1} \rightarrow 0$. ■ (At $t = 50$, $\Sigma_{50} = 1.96$.) Precision grows linearly in the number of tests.

Verification. `ls_ch02.ipynb`, §2.19: the recursion reproduces both closed forms on every path. ✓

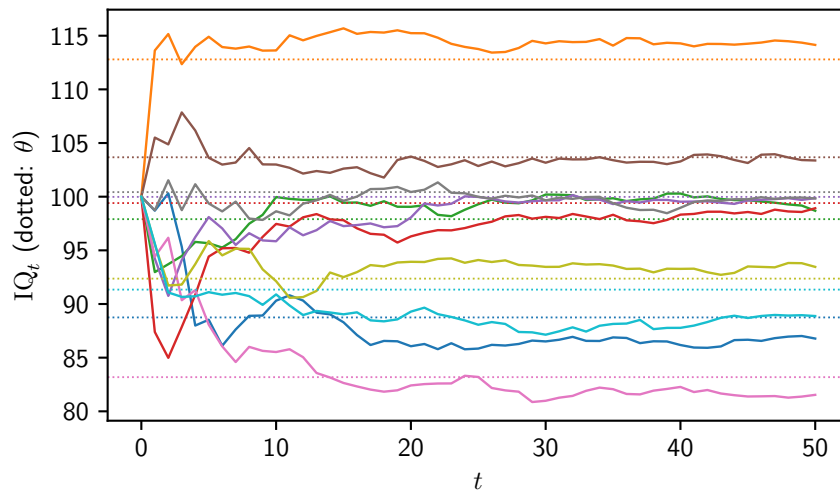


Figure 4: Exercise 2.19: IQ_t for ten draws of θ (dotted lines).

Exercise 2.26 Math and Verbal IQ's

Question

An infinitely lived person's 'true intelligence' θ has two components, math ability θ_1 and verbal ability θ_2 , where $\theta \sim \mathcal{N}\left(\begin{bmatrix} 100 \\ 100 \end{bmatrix}, \begin{bmatrix} 100 & 0 \\ 0 & 100 \end{bmatrix}\right)$. For each date $t \geq 0$, the person takes a single 'test' with the outcome being a univariate random variable $y_t = G_t\theta + v_t$, where v_t is an iid process with distribution $\mathcal{N}(0, 50)$ and $G_t = [.9 \ .1]$ for $t = 0, 2, 4, \dots$ and $G_t = [.01 \ .99]$ for $t = 1, 3, 5, \dots$. Here the person takes a math test at t even and a verbal test at t odd (but you have to know how to read English to survive the math test, and you have to know how to tell time in order to plan your time allocation well for the verbal test). The person's initial IQ vector is $IQ_0 = \begin{bmatrix} 100 \\ 100 \end{bmatrix}$ and at date $t \geq 1$ before the date t test is taken it is $IQ_t = E\theta|y^{t-1}$, where y^{t-1} is the history of test scores from date 0 until date $t-1$.

- (a) Give a recursive formula for IQ_t and for $E(IQ_t - \theta)(IQ_t - \theta)'$.
- (b) Use Matlab to simulate 10 draws of θ and associated paths of y_t, IQ_t for $t = 0, \dots, 50$.
- (c) Show computationally or analytically that $\lim_{t \rightarrow +\infty} E(IQ_t - \theta)(IQ_t - \theta)' = \begin{bmatrix} 0 & 0 \\ 0 & 0 \end{bmatrix}$.

Solution:

Kalman filter with $A = I_2$, $C = 0$, $R = 50$ and a **time-varying** G_t ; $\Sigma_0 = 100I$.

- (a) With $G_t\Sigma_t G_t' + 50$ a scalar,

$$K_t = \frac{\Sigma_t G_t'}{G_t \Sigma_t G_t' + 50}, \quad IQ_{t+1} = IQ_t + K_t (y_t - G_t IQ_t),$$

$$\Sigma_{t+1} = \Sigma_t - \frac{\Sigma_t G_t' G_t \Sigma_t}{G_t \Sigma_t G_t' + 50},$$

and $E(IQ_t - \theta)(IQ_t - \theta)' = \Sigma_t$. (This is (2.7.14) with $A = I$, $C = 0$: the $(A - KG)\Sigma(A - KG)' + K R K'$ form simplifies to the one shown.)

- (b) Figure 5.
- (c) By the matrix inversion lemma the covariance update is, in precision form,

$$\Sigma_{t+1}^{-1} = \Sigma_t^{-1} + \frac{G_t' G_t}{50} \quad \Rightarrow \quad \Sigma_t^{-1} = \frac{I}{100} + \frac{1}{50} \sum_{s=0}^{t-1} G_s' G_s.$$

Each pair of periods adds

$$M = \frac{1}{50} \left(\begin{bmatrix} .81 & .09 \\ .09 & .01 \end{bmatrix} + \begin{bmatrix} .0001 & .0099 \\ .0099 & .9801 \end{bmatrix} \right) = \begin{bmatrix} .016202 & .001998 \\ .001998 & .019802 \end{bmatrix},$$

positive definite (eigenvalues .015313 and .020691). Hence $\lambda_{\min}(\Sigma_t^{-1}) \geq \lfloor t/2 \rfloor (.015313) \rightarrow \infty$ and $\|\Sigma_t\| = 1/\lambda_{\min}(\Sigma_t^{-1}) \rightarrow 0$. ■ Numbers: $\Sigma_{50} = \begin{bmatrix} 2.438 & -.241 \\ -.241 & 2.004 \end{bmatrix}$, $\Sigma_{5000} = \begin{bmatrix} .0250 & -.0025 \\ -.0025 & .0205 \end{bmatrix}$.

Reading. Each test alone is rank-one information, so repeating only the math test would leave the direction orthogonal to $[.9, .1]$ unlearned: Σ_t would not vanish. It is the **alternation** of two linearly independent tests that identifies both abilities. The negative off-diagonal arises because each score loads positively on both abilities: a high math score is partly attributed to verbal ability, so the two estimation errors move in opposite directions.

Verification. `ls_ch02.ipynb`, §2.26: the recursion reproduces the precision form at $t = 50$. ✓

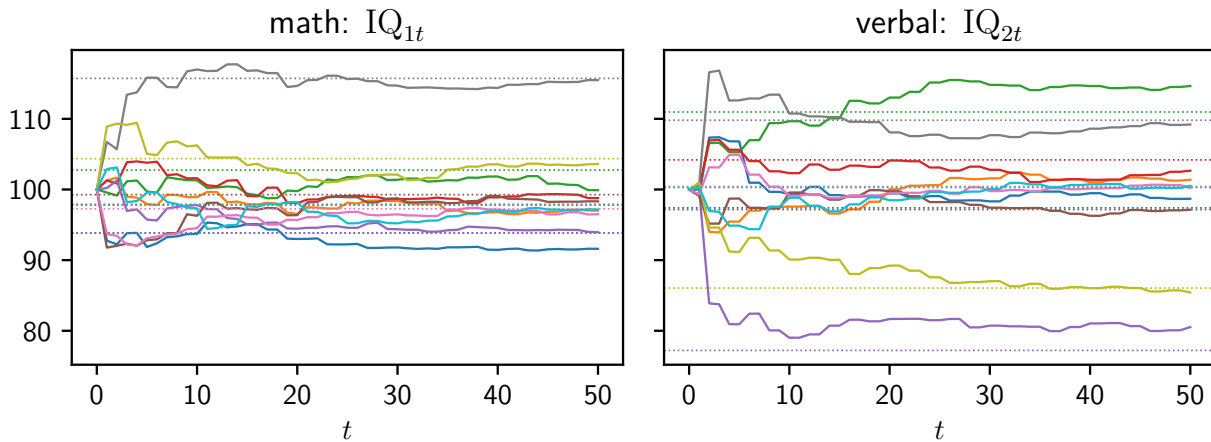


Figure 5: Exercise 2.26: math and verbal IQ for ten draws of θ (dotted: true values).

Exercise 2.20 Random Walk Seen Through Two Signals

Question

A scalar process x_t follows the process $x_{t+1} = x_t + w_{t+1}$ where w is an iid $\mathcal{N}(0, 1)$ scalar process and $x_0 \sim \mathcal{N}(\hat{x}_0, \Sigma_0)$. Each period, an observer receives two signals in the form of a 2×1 vector y_t that obeys $y_t = \begin{bmatrix} 1 \\ 1 \end{bmatrix} x_t + v_t$ where the 2×1 process v_t is iid with distribution $v_t \sim \mathcal{N}(0, R)$ where $R = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$.

- Suppose that $\Sigma_0 = 1.36602540378444$. For $t \geq 0$, find formulas for $E[x_t | y^{t-1}]$, where y^{t-1} is the history of y_s for s from 0 to $t-1$.
- Verify numerically that the matrix $A - KG$ in formula (2.9.3) is stable.
- Find an infinite-order vector autoregression for y_t .

Solution:

$A = 1$, $C = 1$, $G = \mathbf{1} = [1, 1]'$, $R = I_2$.

- (a) *Riccati map.* By Sherman–Morrison, $(\Sigma \mathbf{1}\mathbf{1}' + I)^{-1} = I - \frac{\Sigma}{1+2\Sigma} \mathbf{1}\mathbf{1}'$, so $\mathbf{1}'(\Sigma \mathbf{1}\mathbf{1}' + I)^{-1} \mathbf{1} = 2 - \frac{4\Sigma}{1+2\Sigma} = \frac{2}{1+2\Sigma}$. (2.7.15) becomes

$$\Sigma_{t+1} = \Sigma_t + 1 - \frac{2\Sigma_t^2}{1 + 2\Sigma_t} = \frac{3\Sigma_t + 1}{2\Sigma_t + 1}.$$

Fixed point: $\Sigma(2\Sigma + 1) = 3\Sigma + 1 \Rightarrow 2\Sigma^2 - 2\Sigma - 1 = 0 \Rightarrow \Sigma = \frac{1+\sqrt{3}}{2} = 1.36602540378444$: the given Σ_0 is the fixed point, so $\Sigma_t = \Sigma_0$ and the filter is time invariant from $t = 0$. *Gain.* $K = \Sigma \mathbf{1}'(\Sigma \mathbf{1}\mathbf{1}' + I)^{-1} = \frac{\Sigma}{1+2\Sigma} \mathbf{1}'$; with $1 + 2\Sigma = 2 + \sqrt{3}$, $\frac{\Sigma}{1+2\Sigma} = \frac{(1+\sqrt{3})(2-\sqrt{3})}{2} = \frac{\sqrt{3}-1}{2} \equiv k = .366025$. Hence, with $1 - 2k = 2 - \sqrt{3} = .267949$,

$$\hat{x}_{t+1} = (2 - \sqrt{3}) \hat{x}_t + k(y_{1t} + y_{2t}),$$

$$E[x_t | y^{t-1}] = \hat{x}_t = (2 - \sqrt{3})^t \hat{x}_0 + k \sum_{j=0}^{t-1} (2 - \sqrt{3})^j (y_{1,t-1-j} + y_{2,t-1-j}).$$

The two signals are symmetric, so only their sum matters.

- (b) $A - KG = 1 - 2k = 2 - \sqrt{3} = .2679$, inside the unit circle. ✓
 (c) From (2.9.3), $y_t = G \sum_{j \geq 0} (A - KG)^j K y_{t-j-1} + a_t$:

$$y_t = \begin{bmatrix} 1 \\ 1 \end{bmatrix} k \sum_{j=0}^{\infty} (2 - \sqrt{3})^j \begin{bmatrix} 1 & 1 \end{bmatrix} y_{t-1-j} + a_t,$$

$$E a_t a_t' = \Sigma \mathbf{1}\mathbf{1}' + I = \begin{bmatrix} 2.366025 & 1.366025 \\ 1.366025 & 2.366025 \end{bmatrix}.$$

Each equation has the same coefficients, $A_j = k(2 - \sqrt{3})^{j-1} \mathbf{1}\mathbf{1}'$, $j \geq 1$.

Verification. `ls_ch02.ipynb`, § 2.20: Riccati iteration converges to $(1 + \sqrt{3})/2$; one step from Σ_0 leaves it unchanged; in a 200,000-period simulation the innovations have sample covariance within .05 of $G\Sigma G' + R$ and lag-1 autocovariances below .01. ✓

Exercise 2.21 Impulse Response for a VAR

Question

Find the impulse response function for the state space representation (2.9.1) associated with a vector autoregression.

Solution:

(2.9.1): $\hat{x}_{t+1} = A\hat{x}_t + Ka_t$, $y_t = G\hat{x}_t + a_t$, $Ea_t a_t' = G\Sigma G' + R \equiv \Omega$. Iterating the state equation backward, $\hat{x}_t = \sum_{j \geq 1} A^{j-1} K a_{t-j}$ (plus $A^t \hat{x}_0$ from a finite start), so

$$y_t = a_t + \sum_{j=1}^{\infty} G A^{j-1} K a_{t-j}.$$

$$\frac{\partial y_{t+j}}{\partial a_t'} = \Psi_j = \begin{cases} I_m, & j = 0 \\ G A^{j-1} K, & j \geq 1 \end{cases}$$

Equivalently, the VAR $y_t = \sum_{j \geq 1} \Phi_j y_{t-j} + a_t$ with $\Phi_j = G(A - KG)^{j-1}K$ (2.9.3) inverts to the same Ψ via $\Psi_0 = I$, $\Psi_j = \sum_{k=1}^j \Phi_k \Psi_{j-k}$: the two recursions describe the same linear system ($\hat{x}_{t+1} = (A - KG)\hat{x}_t + Ky_t$ is (2.9.1a) with $a_t = y_t - G\hat{x}_t$ substituted). Because the a_t 's are contemporaneously correlated, the usual practice is to report responses to orthonormalised shocks $\Psi_j L$, with $\Omega = LL'$ (Cholesky); that step imposes a recursive ordering and is not implied by (2.9.1). When A has a unit eigenvalue (Exercise 2.20), Ψ_j need not die out: there $\Psi_j = k\mathbf{1}\mathbf{1}'$ for all $j \geq 1$ (a shock to the random-walk estimate is permanent).

Verification. `ls_ch02.ipynb`, §2.21: $GA^{j-1}K$ equals the VAR inversion for $j < 15$ on the system of Exercise 2.20. ✓

Exercise 2.22 Kalman Filter with Cross-Products

Question

Consider the state space system $x_{t+1} = Ax_t + Cw_{t+1}$, $y_{t+1} = Gx_t + Dw_{t+1}$ where x_t is an $n \times 1$ state vector w_{t+1} is an $m \times 1$ iid process with distribution $\mathcal{N}(0, I)$, y_t is an $m \times 1$ vector of observed variables, and $x_0 \sim \mathcal{N}(\hat{x}_0, \Sigma_0)$. For $t \geq 1$, $\hat{x}_t = E[x_t | y^t]$ where $y^t = [y_t, \dots, y_1]$ and $\Sigma_t = E(x_t - \hat{x}_t)(x_t - \hat{x}_t)'$.

- Show how to select w_{t+1} , C , and D so that Cw_{t+1} and Dw_{t+1} are mutually uncorrelated processes. Also give an example in which Cw_{t+1} and Dw_{t+1} are correlated.
- Construct a recursive representation for $\hat{x}_t$ of the form $\hat{x}_{t+1} = A\hat{x}_t + K_t a_{t+1}$, $y_{t+1} = G\hat{x}_t + a_{t+1}$ where $a_{t+1} = y_{t+1} - E[y_{t+1} | y^t]$ for $t \geq 0$ and verify that $K_t = (CD' + A\Sigma_t G')(DD' + G\Sigma_t G')^{-1}$, $\Sigma_{t+1} = (A - K_t G)\Sigma_t(A - K_t G)' + (C - K_t D)(C - K_t D)'$ and $Ea_{t+1} a_{t+1}' = G\Sigma_t G' + DD'$. *Hint:* apply the population regression formula.

Solution:

- $E(Cw_{t+1})(Dw_{t+1})' = CD'$, so the two are uncorrelated iff $CD' = 0$. Partition $w_{t+1} = [w_{t+1}^x; w_{t+1}^y]$ into independent blocks and set $C = [C_1 \ 0]$, $D = [0 \ D_2]$: then $CD' = C_1 \cdot 0 + 0 \cdot D_2' = 0$ (w then has as many entries as the two blocks together). This recovers the standard system (2.7.1)–(2.7.2) with $R = D_2 D_2'$, up to timing. *Correlated example:* $n = m = 1$, $C = D = 1$: $x_{t+1} = ax_t + w_{t+1}$, $y_{t+1} = gx_t + w_{t+1}$, so $CD' = 1$: the same shock moves the state and the signal. More generally any C, D with $CD' \neq 0$.

- (b) Let $e_t = x_t - \hat{x}_t$; recall $E[e_t | y^t] = 0$ and $w_{t+1} \perp (x_t, y^t)$. *Step 1 (one-step predictions)*. $E[x_{t+1} | y^t] = A\hat{x}_t$ and $E[y_{t+1} | y^t] = G\hat{x}_t$, so

$$a_{t+1} = y_{t+1} - G\hat{x}_t = Ge_t + Dw_{t+1}, \quad x_{t+1} - A\hat{x}_t = Ae_t + Cw_{t+1}.$$

Step 2 (update). y^{t+1} and (y^t, a_{t+1}) span the same space and $a_{t+1} \perp y^t$, so (with normality) $\hat{x}_{t+1} = E[x_{t+1} | y^t] + K_t a_{t+1} = A\hat{x}_t + K_t a_{t+1}$, where K_t is the population regression coefficient of $x_{t+1} - A\hat{x}_t$ on a_{t+1} . Together with Step 1 this gives the two-equation representation. *Step 3 (moments)*. Using $Ee_t w'_{t+1} = 0$ and $Ew_{t+1} w'_{t+1} = I$:

$$E(Ae_t + Cw_{t+1})(Ge_t + Dw_{t+1})' = A\Sigma_t G' + CD', \quad \boxed{Ea_{t+1} a'_{t+1} = G\Sigma_t G' + DD'},$$

$$\boxed{K_t = (CD' + A\Sigma_t G')(DD' + G\Sigma_t G')^{-1}}.$$

Step 4 (error covariance). $e_{t+1} = x_{t+1} - \hat{x}_{t+1} = Ae_t + Cw_{t+1} - K_t(Ge_t + Dw_{t+1}) = (A - K_t G)e_t + (C - K_t D)w_{t+1}$, whose two terms are uncorrelated, so

$$\boxed{\Sigma_{t+1} = (A - K_t G)\Sigma_t(A - K_t G)' + (C - K_t D)(C - K_t D)'}$$

Consistency. $Ee_{t+1} a'_{t+1} = (A - K_t G)\Sigma_t G' + (C - K_t D)D' = A\Sigma_t G' + CD' - K_t(G\Sigma_t G' + DD') = 0$ by the formula for K_t : the new error is orthogonal to the innovation, as a regression residual must be. With $CD' = 0$ and $DD' = R$ the formulas collapse to (2.7.14).

Verification. `ls_ch02.ipynb`, § 2.22: for a random 2×2 system with $CD' \neq 0$, Σ_6 and $\hat{x}_6$ from the recursion equal the brute-force conditional moments of x_6 given $(y_1, \dots, y_6)$ computed from the joint Gaussian covariance. ✓

Exercise 2.24 Stationarity

Question

A pair of scalar stochastic processes (z_t, y_t) evolves according to the state system for $t \geq 0$: $z_{t+1} = .9z_t + w_{t+1}$, $y_t = z_t + v_t$ where w_{t+1} and v_t are mutually uncorrelated scalar Gaussian random variables with means of 0 and variances of 1. Furthermore, $Ew_{t+1}v_s = 0$ for all t, s pairs. In addition, $z_0 \sim \mathcal{N}(\hat{z}_0, \Sigma_0)$.

- Is $\{z_t\}$ Markov? Explain.
- Is $\{y_t\}$ Markov? Explain.
- Define what it would mean for the scalar process $\{z_t\}$ to be *covariance stationary*.
- Find values of $(\hat{z}_0, \Sigma_0)$ that make the process for $\{z_t\}$ covariance stationary.
- Assume that y_t is observable, but that z_t is not. Define what it would mean for the scalar process y_t to be *covariance stationary*.
- Describe in as much detail as you can how to represent the distribution of y_t conditional on the infinite history y^{t-1} in the form $y_t \sim \mathcal{N}(E[y_t | y^{t-1}], \Omega_t)$.

Solution:

- (a) Yes. $z_{t+1}|z^t \sim \mathcal{N}(.9z_t, 1)$ because w_{t+1} is independent of z^t : the conditional distribution depends on the past only through z_t .
- (b) No. y_t is a noisy reading of z_t , and past y 's all carry information about z_t beyond y_t . Formally $(1 - .9L)y_t = w_t + v_t - .9v_{t-1}$ is an ARMA(1,1), and (part f) $E[y_{t+1}|y^t] = .537667 \sum_{j \geq 0} (.362333)^j y_{t-j}$ puts weight on every lag, so y is not Markov of any finite order.
- (c) $Ez_t = \mu$ for all t , and $E(z_t - \mu)(z_{t-j} - \mu) = C_z(j)$ depends only on j , not on t .
- (d) $Ez_t = .9^t \hat{z}_0$ is constant iff $\hat{z}_0 = 0$. $\text{Var}z_{t+1} = .81\text{Var}z_t + 1$ is constant iff $\Sigma_0 = \frac{1}{1-.81} = 5.263158$. So $(\hat{z}_0, \Sigma_0) = (0, 1/.19)$; then $C_z(j) = .9^j/.19$.
- (e) Same definition for y : $Ey_t = \mu_y$ for all t , and $E(y_t - \mu_y)(y_{t-j} - \mu_y) = C_y(j)$ depends only on j . Under (d), $C_y(0) = \frac{1}{.19} + 1$ and $C_y(j) = \frac{.9^j}{.19}$ for $j \geq 1$; covariance stationarity of y is a statement about observables only.
- (f) Run the time-invariant Kalman filter (section 2.9), which conditions on the semi-infinite past: $A = .9$, $C = 1$, $G = 1$, $R = 1$. The Riccati equation $\Sigma = .81\Sigma + 1 - \frac{.81\Sigma^2}{\Sigma+1}$, multiplied through by $\Sigma+1$, becomes $\Sigma^2 + \Sigma = .81\Sigma^2 + .81\Sigma + \Sigma + 1 - .81\Sigma^2$, i.e.

$$\Sigma^2 - .81\Sigma - 1 = 0 \Rightarrow \Sigma = \frac{.81 + \sqrt{.6561 + 4}}{2} = \frac{.81 + 2.157800}{2} = 1.483900.$$

Then $K = \frac{.9\Sigma}{\Sigma+1} = \frac{1.335510}{2.483900} = .537667$, $A - KG = .362333$, and

$$\boxed{\begin{aligned} y_t | y^{t-1} &\sim \mathcal{N}(\hat{z}_t, \Omega), & \Omega &= \Sigma + 1 = 2.483900, \\ \hat{z}_t &= .537667 \sum_{j=0}^{\infty} (.362333)^j y_{t-1-j} \end{aligned}}$$

recursively $\hat{z}_{t+1} = .9\hat{z}_t + K(y_t - \hat{z}_t)$. With the infinite history $\Omega_t = \Omega$ is constant; from a finite start, $\Omega_t = \Sigma_t + 1$ with Σ_t from (2.7.15), converging to Ω .

Verification. `ls_ch02.ipynb`, §2.24: Ω equals the Kolmogorov–Szegő one-step prediction variance $\exp\{\frac{1}{2\pi} \int \log S_y\}$ with $S_y(\omega) = |1 - .9e^{-i\omega}|^{-2} + 1$, to 6 digits. ✓

Exercise 2.28 Invertibility

Question

A univariate stochastic process y_t has a first-order moving average representation $y_t = \epsilon_t - 2\epsilon_{t-1}$ (1) where $\{\epsilon_t\}$ is an i.i.d. process distributed $\mathcal{N}(0, 1)$.

(a) Argue that ϵ_t cannot be expressed as linear combination of $y_{t-j}, j \geq 0$ where the sum of the squares of the weights is finite. This means that ϵ_t is not in the space spanned by square summable linear combinations of the infinite history y^t .

(b) Write equation (1) as a state space system, indicating the matrices A, C, G .

(c) Using the matlab program `kfilter.m` to compute an innovations representation for $\{y_t\}$. Verify that the innovations representation for y_t can be represented as $y_t = a_t - .5a_{t-1}$ (2) where $a_t = y_t - E[y_t|y^{t-1}]$ is a serially uncorrelated process. Compute the variance of a_t . Is it larger or smaller than the variance of ϵ_t ?

(d) Find an autoregressive representation for y_t of the form $y_t = \sum_{j=1}^{\infty} A_j y_{t-j} + a_t$ (3) where $E a_t y_{t-j} = 0$ for $j \geq 1$. (*Hint*: either use formula (2) or else remember formula (2.9.3).)

(e) Is y_t Markov? Is $[y_t \ y_{t-1}]'$ Markov? Is $[y_t \ y_{t-1} \ \cdots \ y_{t-10}]'$ Markov?

(f) Extra credit. Verify that ϵ_t can be expressed as a square summable linear combination of $y_{t+j}, j \geq 1$.

Solution:

(a) Suppose $\epsilon_t = \sum_{j \geq 0} h_j y_{t-j}$. Substituting (1), $\sum_j h_j (\epsilon_{t-j} - 2\epsilon_{t-j-1}) = \epsilon_t$. The ϵ 's are orthonormal, so coefficients must match: ϵ_t : $h_0 = 1$; $\epsilon_{t-k}, k \geq 1$: $h_k - 2h_{k-1} = 0$. Hence $h_k = 2^k$, i.e. $h(L) = (1 - 2L)^{-1}$, and $\sum_k h_k^2 = \sum 4^k = \infty$. No square-summable combination of y^t reproduces ϵ_t . The root of $1 - 2z$ is $z = \frac{1}{2}$, inside the unit circle: (1) is a non-fundamental representation.

(b) $x_t = \begin{bmatrix} \epsilon_t \\ \epsilon_{t-1} \end{bmatrix}$, $A = \begin{bmatrix} 0 & 0 \\ 1 & 0 \end{bmatrix}$, $C = \begin{bmatrix} 1 \\ 0 \end{bmatrix}$, $G = \begin{bmatrix} 1 & -2 \end{bmatrix}$, and no measurement error ($R = 0$): $x_{t+1} = Ax_t + C\epsilon_{t+1}$, $y_t = Gx_t$.

(c) The steady state of (2.7.14) is $\Sigma = \begin{bmatrix} 1 & 0 \\ 0 & .75 \end{bmatrix}$, $K = \begin{bmatrix} 0 \\ .25 \end{bmatrix}$. Check: $A\Sigma G' = A[1, -1.5]' = [0, 1]'$ and $G\Sigma G' = 1 + 4(.75) = 4$, so $K = [0, .25]'$; $A - KG = \begin{bmatrix} 0 & 0 \\ .75 & .5 \end{bmatrix}$ and $(A - KG)\Sigma(A - KG)' + CC' = \begin{bmatrix} 1 & 0 \\ 0 & .5625 + .1875 \end{bmatrix} = \Sigma \checkmark$. The innovations representation: $\hat{x}_t = [0, .25a_{t-1}]'$ (nobody can forecast ϵ_t), so $y_t = G\hat{x}_t + a_t = a_t - 2(.25)a_{t-1} = a_t - .5a_{t-1}$, with $\text{Var}(a_t) = G\Sigma G' = 4$, **larger** than $\text{Var}(\epsilon_t) = 1$. Autocovariance check: $\text{Vary} = 1 + 4 = 5 = 4(1 + .25)$ and $\text{Cov}(y_t, y_{t-1}) = -2 = -.5 \times 4 \checkmark$. a_t is the one-step error from forecasting with the past only; ϵ_t would need the future (part f), so it is harder to forecast and the innovation variance is larger. ($\Sigma_{22} = .75$: even after seeing y_{t-1} , ϵ_{t-1} remains uncertain.)

(d) $a_t = (1 - .5L)^{-1}y_t = \sum_{j \geq 0} .5^j y_{t-j}$, so

$$y_t = - \sum_{j=1}^{\infty} .5^j y_{t-j} + a_t, \quad A_j = -.5^j.$$

Via (2.9.3): $(A - KG)^{j-1}K = [0, .25(.5)^{j-1}]'$, so $A_j = G(A - KG)^{j-1}K = -2(.25)(.5)^{j-1} = -.5^j \checkmark$. $Ea_t y_{t-j} = 0$ because a_t is the innovation.

(e) No to all three. Gaussian conditional distributions are pinned down by the conditional mean and variance, and $E[y_{t+1}|y^t] = -\sum_{j \geq 0} .5^{j+1} y_{t-j}$ puts nonzero weight on every lag, so no finite window $[y_t, \dots, y_{t-k}]$ is a sufficient statistic. The pair $[y_t, a_t]'$ (or $\hat{x}_t$) is Markov.

(f) Solve (1) forward: $\epsilon_t = \frac{\epsilon_{t+1} - y_{t+1}}{2}$. Iterating, $\epsilon_t = - \sum_{j=1}^{\infty} .5^j y_{t+j}$, square summable.

Check: $\sum_{j \geq 1} .5^j (\epsilon_{t+j} - 2\epsilon_{t+j-1}) = \sum_{j \geq 1} .5^j \epsilon_{t+j} - \sum_{k \geq 0} .5^k \epsilon_{t+k} = -\epsilon_t \checkmark$.

Verification. `1s_ch02.ipynb`, § 2.28: Riccati iteration gives $K, \Sigma, \Omega = 4$; OLS of y_t on 20 lags in a 200,000-period sample gives $(-.498, -.246, -.123, \dots)$ and residual variance 4.02; part (f) reconstructs ϵ_t to 2×10^{-15} . $\checkmark$

Exercise 2.23 A Monopolist, Learning, and Ergodicity

Question

A monopolist produces a quantity Q_t of a single good in every period $t \geq 0$ at zero cost. At the beginning of each period $t \geq 0$, before output price p_t is observed, the monopolist sets quantity Q_t to maximize $E_{t-1} p_t Q_t$ (1) where p_t satisfies the linear inverse demand curve $p_t = a - bQ_t + \sigma_p \epsilon_t$ (2) where $b > 0$ is a constant known to the firm, ϵ_t is an i.i.d. scalar with distribution $\epsilon_t \sim \mathcal{N}(0, 1)$, and the constant in the inverse demand curve a is a scalar random variable unknown to the firm and whose unconditional distribution is $a \sim \mathcal{N}(\mu_a, \sigma_a^2)$, where $\mu_a > 0$ is large relative to $\sigma_a > 0$. Assume that the random variable a is independent of ϵ_t for all t . Before the firm chooses Q_0 , it knows the unconditional distribution of a , but not the realized value of a . For each $t \geq 0$, the firm wants to estimate a because it wants to make a good decision about output Q_t . At the end of each period t , when it must set Q_{t+1} , the firm observes p_t and also of course knows the value of Q_t that it had set. In (1), for $t \geq 1$, $E_{t-1}(\cdot)$ denotes the mathematical expectation conditional on the history of signals $p_s, q_s, s = 0, \dots, t-1$; for $t = 0$, $E_{-1}(\cdot)$ denotes the expectation conditioned on no previous observations of p_t, Q_t .

- (a) What is the optimal setting for Q_0 ? For each date $t \geq 0$, determine the firm's optimal setting for Q_t as a function of the information p^{t-1}, Q^{t-1} that the firm has when it sets Q_t .
- (b) Under the firm's optimal policy, is the pair (p_t, Q_t) Markov?
- (c) 'Finding the state is an art.' Find a recursive representation of the firm's optimal policy for setting Q_t for $t \geq 0$. Interpret the state variables that you propose.
- (d) Under the firm's optimal rule for setting Q_t , does the random variable $E_{t-1} p_t$ converge to a constant as $t \rightarrow +\infty$? If so, prove that it does and find the limiting value. If not, tell why it does not converge.
- (e) Now suppose that instead of maximizing (1) each period, there is a single infinitely lived monopolist who once and for all before time 0 chooses a plan for an entire sequence $\{Q_t\}_{t=0}^{\infty}$, where the Q_t component has to be a measurable function of (p^{t-1}, q^{t-1}) , and where the monopolist's objective is to maximize $E_{-1} \sum_{t=0}^{\infty} \beta^t p_t Q_t$ (3) where $\beta \in (0, 1)$ and E_{-1} denotes the mathematical expectation conditioned on the null history. Get as far as you can in deriving the monopolist's optimal sequence of decision rules.

Solution:

Key observation. Since Q_t is known, the firm can form the signal

$$s_t \equiv p_t + bQ_t = a + \sigma_p \epsilon_t,$$

a noisy reading of a whose distribution does not depend on Q_t . This is Jovanovic's signal-extraction problem (2.10.8) with $R = \sigma_p^2$ and prior $(m_{-1}, \Sigma_0) = (\mu_a, \sigma_a^2)$.

- (a) Q_t is chosen before p_t is seen, so $E_{t-1} p_t Q_t = Q_t (E_{t-1} a - bQ_t)$. The first-order

condition $E_{t-1}a - 2bQ_t = 0$ gives

$$\boxed{Q_t = \frac{m_{t-1}}{2b}}, \quad m_{t-1} \equiv E[a|p^{t-1}, Q^{t-1}], \quad \boxed{Q_0 = \frac{\mu_a}{2b}}.$$

Normal-normal updating with t signals gives, explicitly in the data,

$$m_{t-1} = \frac{\mu_a/\sigma_a^2 + \sum_{s=0}^{t-1}(p_s + bQ_s)/\sigma_p^2}{1/\sigma_a^2 + t/\sigma_p^2}, \quad Q_t = \frac{\mu_a/\sigma_a^2 + \sum_{s=0}^{t-1}(p_s + bQ_s)/\sigma_p^2}{2b(1/\sigma_a^2 + t/\sigma_p^2)}.$$

- (b) From (2.10.8), $m_t = (1 - K_t)m_{t-1} + K_t s_t$ with the *deterministic* gain $K_t = \Sigma_t/(\Sigma_t + \sigma_p^2)$. Substituting $m_{t-1} = 2bQ_t$ and $s_t = p_t + bQ_t$,

$$Q_{t+1} = \frac{(1 - K_t)2bQ_t + K_t(p_t + bQ_t)}{2b},$$

a function of (p_t, Q_t) and of t only. Under the firm's predictive distribution, $a|\text{history} \sim \mathcal{N}(m_t, \Sigma_{t+1})$ depends only on $m_t = 2bQ_{t+1}$, so the law of (p_{t+1}, Q_{t+1}) given the whole past depends only on (p_t, Q_t) and t . Answer: **yes, but not time-homogeneous**: the transition law changes with t through K_t, Σ_t . Adding Σ_t (or t) to the pair gives a time-invariant Markov state.

- (c) State (m_{t-1}, Σ_t) , initialised at (μ_a, σ_a^2) :

$$Q_t = \frac{m_{t-1}}{2b}; \quad \text{observe } p_t; \quad K_t = \frac{\Sigma_t}{\Sigma_t + \sigma_p^2},$$

$$m_t = m_{t-1} + K_t(p_t + bQ_t - m_{t-1}), \quad \Sigma_{t+1} = \frac{\Sigma_t \sigma_p^2}{\Sigma_t + \sigma_p^2}.$$

m_{t-1} is the firm's current estimate of the demand intercept (it alone drives output); Σ_t is its uncertainty, with precision $1/\Sigma_t = 1/\sigma_a^2 + t/\sigma_p^2$ growing linearly, and it governs only how strongly the next surprise revises the estimate.

- (d) $E_{t-1}p_t = m_{t-1} - bQ_t = \frac{m_{t-1}}{2}$. From the explicit formula,

$$m_{t-1} - a = \frac{(\mu_a - a)/\sigma_a^2 + \sigma_p^{-1} \sum_{s < t} \epsilon_s}{1/\sigma_a^2 + t/\sigma_p^2} = \frac{(\mu_a - a)/(t\sigma_a^2) + \sigma_p^{-1} \frac{1}{t} \sum_{s < t} \epsilon_s}{1/(t\sigma_a^2) + 1/\sigma_p^2} \rightarrow 0$$

almost surely by the strong law of large numbers. Hence $\boxed{E_{t-1}p_t \rightarrow a/2}$ (and $Q_t \rightarrow a/(2b)$, the full-information monopoly output). It converges, but to the **random variable** $a/2$, not to a constant common to all histories: each sample path settles at the value of its own a . The process is not ergodic, since time averages recover the realised a , not the population mean μ_a .

- (e) The information is **independent of the plan**: whatever measurable plan is chosen, the firm can recover $s_s = p_s + bQ_s$ and conversely $p_s = s_s - bQ_s$, so

(p^{t-1}, Q^{t-1}) and s^{t-1} carry the same information about a . By the law of iterated expectations and Q_t being known at $t-1$,

$$E_{-1} \sum_t \beta^t p_t Q_t = \sum_t \beta^t E_{-1} [Q_t (m_{t-1} - bQ_t)],$$

and the distribution of m_{t-1} does not depend on past Q 's. The problem separates date by date, and each term is maximised pointwise by $Q_t = m_{t-1}/(2b)$. So the **myopic rule of part a is optimal** for the infinitely lived monopolist, for any β . There is no motive to experiment, because output does not change what prices reveal. That would differ if the noise entered multiplicatively with Q_t , or if the unknown parameter were the slope b .

Verification. `1s_ch02.ipynb`, §2.23 ($\mu_a = 10, \sigma_a = 1, \sigma_p = 2, b = 1$): the recursion reproduces the explicit formula for m_t and $1/\Sigma_t$ at every date; $E_{t-1}p_t$ settles at each path's $a/2$ (Figure 6). ✓

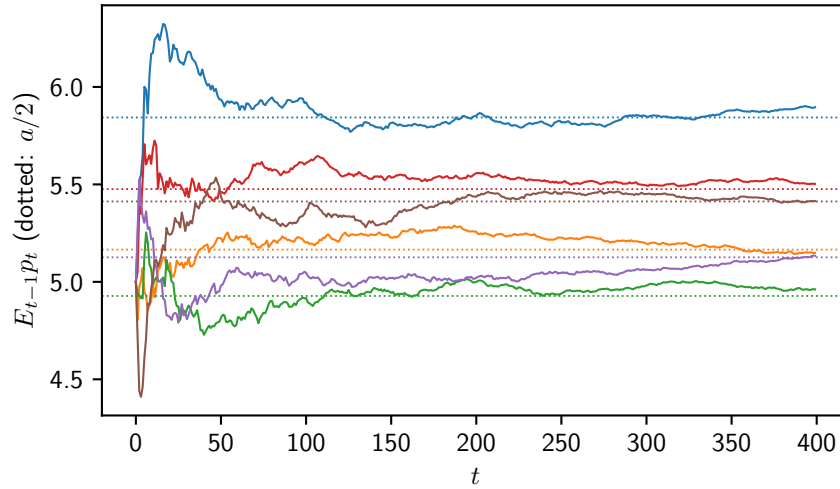


Figure 6: Exercise 2.23: $E_{t-1}p_t$ along six sample paths, each converging to its own $a/2$ (dotted).

The LQ Permanent Income Model and Howard Improvement

Exercise 2.13 Permanent Income by Howard Improvement

Question

A household has preferences over consumption processes $\{c_t\}_{t=0}^{\infty}$ that are ordered by $-.5 \sum_{t=0}^{\infty} \beta^t [(c_t - 30)^2 + .000001b_t^2]$ (1) where $\beta = .95$. The household chooses a consumption, borrowing plan to maximize (1) subject to the sequence of budget constraints $c_t + b_t = \beta b_{t+1} + y_t$ for $t \geq 0$, where b_0 is an initial condition, β^{-1} is the one-period gross risk-free interest rate, b_t is the household's one-period debt that is due in period t , and y_t is its labor income, which obeys the second-order autoregressive process $(1 - \rho_1 L - \rho_2 L^2)y_{t+1} = (1 - \rho_1 - \rho_2)5 + .05w_{t+1}$ where $\rho_1 = 1.3, \rho_2 = -.4$.

- (a) Define the *state* of the household at t as $x_t = [1 \ b_t \ y_t \ y_{t-1}]'$ and the *control* as $u_t = (c_t - 30)$. Then express the transition law facing the household in the form (2.4.22). Compute the eigenvalues of A . Compute the zeros of the characteristic polynomial $(1 - \rho_1 z - \rho_2 z^2)$ and compare them with the eigenvalues of A . (*Hint*: To compute the zeros in Matlab, set $a = [.4 \ -1.3 \ 1]$ and call `roots(a)`. The zeros of $(1 - \rho_1 z - \rho_2 z^2)$ equal the *reciprocals* of the eigenvalues of the associated A .)
- (b) Write a Matlab program that uses the Howard improvement algorithm (2.4.30) to compute the household's optimal decision rule for $u_t = c_t - 30$. Tell how many iterations it takes for this to converge (also tell your convergence criterion).
- (c) Use the household's optimal decision rule to compute the law of motion for x_t under the optimal decision rule in the form $x_{t+1} = (A - BF^*)x_t + Cw_{t+1}$, where $u_t = -F^*x_t$ is the optimal decision rule. Using Matlab, compute the impulse response function of $[c_t \ b_t]'$ to w_{t+1} . Compare these with the theoretical expressions (2.12.18).

Solution:

- (a) From the budget constraint, $b_{t+1} = \beta^{-1}(c_t + b_t - y_t) = \beta^{-1}(30 + b_t - y_t + u_t)$; income is $y_{t+1} = (1 - 1.3 + .4)5 + 1.3y_t - .4y_{t-1} + .05w_{t+1} = .5 + 1.3y_t - .4y_{t-1} + .05w_{t+1}$. So $x_{t+1} = Ax_t + Bu_t + Cw_{t+1}$ with

$$A = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 30/\beta & 1/\beta & -1/\beta & 0 \\ .5 & 0 & 1.3 & -.4 \\ 0 & 0 & 1 & 0 \end{bmatrix}, \quad B = \begin{bmatrix} 0 \\ 1/\beta \\ 0 \\ 0 \end{bmatrix}, \quad C = \begin{bmatrix} 0 \\ 0 \\ .05 \\ 0 \end{bmatrix},$$

and the loss $\sum \beta^t (x_t' R x_t + u_t' Q u_t)$ with $R = 10^{-6} e_2 e_2'$, $Q = 1$ (the factor .5 is irrelevant). A is block triangular, so its eigenvalues are 1 (constant), $1/\beta = 1.052632$ (debt), and the roots of $\lambda^2 - 1.3\lambda + .4 = 0$: $\lambda = \frac{1.3 \pm \sqrt{1.69 - 1.6}}{2} = \frac{1.3 \pm .3}{2} = [.8, .5]$. The zeros of $1 - 1.3z + .4z^2$ are $z = \frac{1.3 \pm .3}{.8} = [2, 1.25]$, the reciprocals of .5 and .8, as the hint says.

- (b) Howard (2.4.30): $P_j = R + F_j' Q F_j + \beta(A - BF_j)' P_j (A - BF_j)$ (a Lyapunov equation in $\sqrt{\beta}(A - BF_j)$, solved by doubling), $F_{j+1} = \beta(Q + \beta B' P_j B)^{-1} B' P_j A$. A is unstable ($1/\beta$), so the initial rule must stabilise $\sqrt{\beta}(A - BF_0)$; we use the permanent-income slope on debt, $F_0 = [0, 1 - \beta, 0, 0]$, which sets the closed-loop

debt root to 1. Convergence criterion $\max |F_{j+1} - F_j| < 10^{-10}$: **4 iterations**, and

$$F^* = [26.230541, .050019, -.396944, .150836],$$

$$c_t = 30 - F^* x_t = 3.7695 - .050019 b_t + .396944 y_t - .150836 y_{t-1}.$$

Compare the PIH rule (2.12.9): coefficient $-(1-\beta) = -.05$ on b_t , and on (y_t, y_{t-1}) , $(1-\beta)U_y(I-\beta A_{22})^{-1}$ with $A_{22} = \begin{bmatrix} 1.3 & -.4 \\ 1 & 0 \end{bmatrix}$: $\det(I-\beta A_{22}) = -.235 + .361 = .126$, first row of the inverse $[1, -.38]/.126 = [7.93651, -3.01587]$, times $.05$: $[\text{.396825}, -.150794]$. The tiny debt penalty moves them only in the fourth digit.

- (c) Closed loop $A - BF^*$ has eigenvalues $.5, .8, .99998, 1$: debt has a near-unit root (it would be exactly 1 without the 10^{-6} penalty). Impulse responses to $w_{t+1} = 1$: c_{t+1+j} responds by $-F^*(A - BF^*)^j C$ and b_{t+1+j} by $e'_2(A - BF^*)^j C$. Theory (2.12.18):

$$d(\beta) = U_y(I - \beta A_{22})^{-1} C_2 = \frac{.05}{1 - 1.3(.95) + .4(.95)^2} = \frac{.05}{.126} = .396825,$$

$$c : (1 - \beta)d(\beta) = \boxed{.019841} \text{ for all } j,$$

$$b_{t+1+j} : U_y(I - \beta A_{22})^{-1} A_{22}^j C_2 - d(\beta) \quad (= 0, -.031746, -.080952, -.132222, \dots).$$

Numerically, c jumps by $.019847$ and stays there (a box), and b follows $0, -.03174, -.08094, -.13220, \dots$, matching theory to about $.03\%$ (Figure 7). A positive income shock raises consumption by the annuity value of the revised income forecast; since the AR(2) makes income rise further before decaying, the household saves (debt falls), and the fall in debt is permanent.

Verification. `ls_ch02.ipynb`, § 2.13: Howard's F^* equals the Riccati solution from `scipy.linalg.solve_discrete_are`; impulse responses agree with (2.12.18) within 2% over 20 periods. ✓

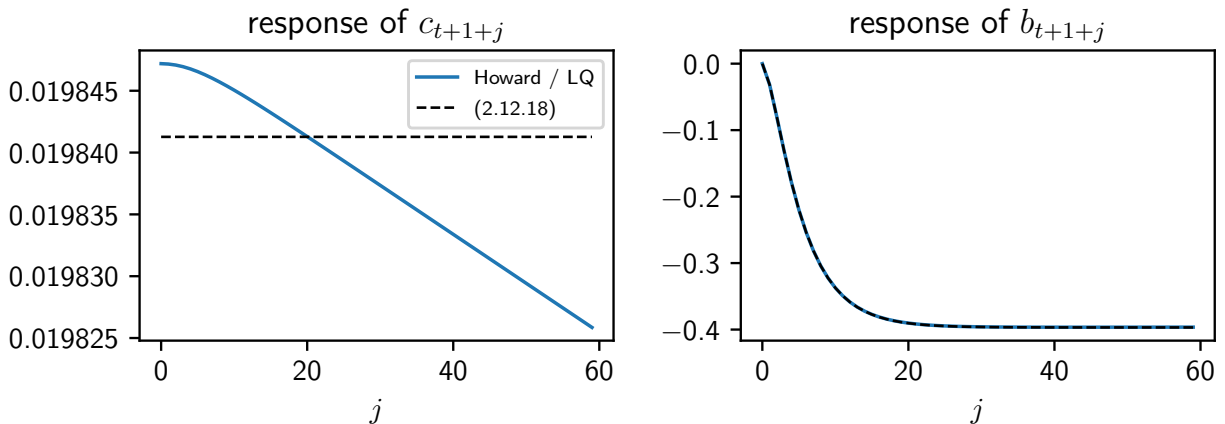


Figure 7: Exercise 2.13: impulse responses of consumption and debt to w_{t+1} , Howard/LQ solution against formula (2.12.18).

Exercise 2.25 Consumption: the Two Classic Examples

Question

- (a) Please use formulas (2.12.18) to verify formulas (2.12.21) and (2.12.23)–(2.12.24) of subsection 2.12.3.
 (b) Please use formulas (2.10.3) to compute the decision rules in formulas (2.12.21) and (2.12.23) for the following parameter values: $\beta = .95, \sigma_1 = \sigma_2 = 1$.
 (c) Please use formulas (2.10.3) to compute the decision rules in formulas (2.12.21) and (2.12.23) for the following parameter values: $\beta = .95, \sigma_1 = 2, \sigma_2 = 1$.
 (d) Please use formula (2.12.20) to confirm formulas (2.12.22) and (2.12.25).

Solution:

(2.12.18a) says $c_{t+1} - c_t = (1 - \beta)U_y(I - \beta A_{22})^{-1}C_2 w_{t+1}$.

- (a) *Example 1:* $A_{22} = \text{diag}(1, 0)$, $C_2 = \text{diag}(\sigma_1, \sigma_2)$, $U_y = [1, 1]$. $(I - \beta A_{22})^{-1} = \text{diag}(\frac{1}{1-\beta}, 1)$, so $U_y(I - \beta A_{22})^{-1}C_2 = [\frac{\sigma_1}{1-\beta}, \sigma_2]$, and

$$c_{t+1} - c_t = (1 - \beta) \left[\frac{\sigma_1}{1-\beta} w_{1t+1} + \sigma_2 w_{2t+1} \right] = \sigma_1 w_{1t+1} + (1 - \beta)\sigma_2 w_{2t+1}. \quad (2.12.21) \checkmark$$

Example 2: $z_t = [y_t, a_t]'$, $A_{22} = \begin{bmatrix} 1 & -(1-K) \\ 0 & 0 \end{bmatrix}$, $C_2 = \begin{bmatrix} 1 \\ 1 \end{bmatrix}$, $U_y = [1, 0]$. $I - \beta A_{22} = \begin{bmatrix} 1-\beta & \beta(1-K) \\ 0 & 1 \end{bmatrix}$, with inverse $\begin{bmatrix} \frac{1}{1-\beta} & -\frac{\beta(1-K)}{1-\beta} \\ 0 & 1 \end{bmatrix}$. Then $U_y(I - \beta A_{22})^{-1}C_2 = \frac{1-\beta(1-K)}{1-\beta}$ and

$$c_{t+1} - c_t = [1 - \beta(1-K)] a_{t+1}. \quad (2.12.23) \checkmark$$

(2.12.18d) row 1: $y_{t+1} = y_t - (1-K)a_t + a_{t+1}$, i.e. $y_{t+1} - y_t = a_{t+1} - (1-K)a_t$ (2.12.24) $\checkmark$.

- (b,c) *Mapping to Muth.* $y_t = z_{1t} + z_{2t}$ with $z_{1,t+1} = z_{1t} + \sigma_1 w_{1,t+1}$ and $z_{2t} = \sigma_2 w_{2t}$ iid: Muth's model (2.10.2) with $Q = \sigma_1^2$, $R = \sigma_2^2$. The fixed point of (2.10.3b), $\Sigma = \Sigma + Q - \frac{\Sigma^2}{\Sigma + R}$, gives $\Sigma^2 = Q(\Sigma + R)$, so

$$\Sigma = \frac{Q + \sqrt{Q^2 + 4QR}}{2}, \quad K = \frac{\Sigma}{\Sigma + R}, \quad \text{Var}(a_t) = \Sigma + R.$$

	Ex. 1: Δc_{t+1}	Σ	K	$\text{Var}(a)$	Ex. 2: Δc_{t+1}	Ex. 2: Δb_{t+1}
(b)	$w_{1t+1} + .05w_{2t+1}$	1.618034	.618034	2.618034	.637132 a_{t+1}	-.381966 a_t
(c)	$2w_{1t+1} + .05w_{2t+1}$	4.828427	.828427	5.828427	.837006 a_{t+1}	-.171573 a_t

Here $\Delta c_{t+1} = c_{t+1} - c_t$ and $\Delta b_{t+1} = b_{t+1} - b_t$; row (b) has $\sigma_1 = \sigma_2 = 1$ and $\Sigma = \frac{1+\sqrt{5}}{2}$, row (c) has $\sigma_1 = 2, \sigma_2 = 1$ and $\Sigma = 2 + 2\sqrt{2}$. Arithmetic for (b):

$K = \frac{1.618034}{2.618034} = .618034$, $1 - \beta(1 - K) = 1 - .95(.381966) = .637132$. For (c): $Q = 4$, $\Sigma^2 - 4\Sigma - 4 = 0 \Rightarrow \Sigma = 2 + 2\sqrt{2}$; $K = \frac{4.828427}{5.828427} = .828427$; $1 - .95(.171573) = .837006$. A larger permanent variance makes the consumer attribute more of each surprise to the permanent component, so consumption responds more and less is saved.

- (d) (2.12.20): $b_{t+1} - b_t = U_y(I - \beta A_{22})^{-1}(A_{22} - I)z_t$. *Example 1:* $\left[\frac{1}{1-\beta}, 1\right] \text{diag}(0, -1)z_t = -z_{2t} = -\sigma_2 w_{2t}$ (2.12.22) ✓. *Example 2:* $A_{22} - I = \begin{bmatrix} 0 & -(1-K) \\ 0 & -1 \end{bmatrix}$, and $\left[\frac{1}{1-\beta}, -\frac{\beta(1-K)}{1-\beta}\right](A_{22}-I) = \left[0, \frac{-(1-K)+\beta(1-K)}{1-\beta}\right] = [0, -(1-K)]$, so $b_{t+1} - b_t = (K-1)a_t$ (2.12.25) ✓.

Verification. `ls_ch02.ipynb`, § 2.25: A_{22}, C_2 run through (2.12.18) numerically; Riccati iteration gives the same K ; the innovations form reproduces the autocovariances of Δy : $\text{Var}(a)(1 + (1-K)^2) = \sigma_1^2 + 2\sigma_2^2$ and $-(1-K)\text{Var}(a) = -\sigma_2^2$. ✓

Exercise 2.27 Permanent Income Model Again

Question

Each of two consumers named $i = 1, 2$ has preferences over consumption streams that are ordered by the utility functional $E_0 \sum_{t=0}^{\infty} \beta^t u(c_t^i)$ (1) where E_t is the mathematical expectation conditioned on the consumer's time t information, c_t^i is time t consumption of consumer i at time t , $u(c)$ is a strictly concave one-period utility function, and $\beta \in (0, 1)$ is a discount factor. The consumer maximizes (1) by choosing a consumption, borrowing plan $\{c_t^i, b_{t+1}^i\}_{t=0}^{\infty}$ subject to the sequence of budget constraints $c_t^i + b_t^i = R^{-1}b_{t+1}^i + y_t^i$ where y_t is an exogenous stationary endowment process, R is a constant gross risk-free interest rate, b_t^i is one-period risk-free debt maturing at t , and $b_0^i = 0$ is a given initial condition. Assume that $R^{-1} = \beta$. We impose the following condition on the consumption, borrowing plan of consumer i : $E_0 \sum_{t=0}^{\infty} \beta^t (b_t^i)^2 < +\infty$. Assume the quadratic utility function $u(c_t) = -.5(c_t - \gamma)^2$, where $\gamma > 0$ is a bliss level of consumption. Negative consumption rates are allowed. Let $s_t \in \{0, 1\}$ be an i.i.d. process with $\text{Prob}(s_t = 1) = \text{Prob}(s_1 = 0) = .5$. The endowment process of consumer 1 is $y_t^1 = 1 - .5s_t$ and the endowment process of person 2 is $y_t^2 = .5 + .5s_t$. Thus, the two consumers' endowment processes are perfectly negatively correlated i.i.d. processes with means of .75.

- (a) Find optimal decision rules for consumption for both consumers. Prove that the consumers' optimal decisions imply the following laws of motion for b_t^1, b_t^2 : $b_{t+1}^1(s_t = 0) = b_t^1 - .25$, $b_{t+1}^1(s_t = 1) = b_t^1 + .25$, $b_{t+1}^2(s_t = 0) = b_t^2 + .25$, $b_{t+1}^2(s_t = 1) = b_t^2 - .25$.
- (b) Show that for each consumer, c_t^i, b_t^i are co-integrated.
- (c) Verify that b_{t+1}^i is risk-free in the sense that conditional on information available at time t , it is independent of news arriving at time $t + 1$.
- (d) Verify that with the initial conditions $b_0^1 = b_0^2 = 0$, the following two equalities obtain: $b_t^1 + b_t^2 = 0 \ \forall t \geq 1$, $c_t^1 + c_t^2 = 1.5 \ \forall t \geq 1$. Use these conditions to interpret the decision rules that you have computed as describing a closed pure consumption loans economy in which consumers 1 and 2 borrow and lend with each other and in which the risk-free asset is a one-period IOU from one of the consumers to the other.
- (e) Define the 'stochastic discount factor of consumer i ' as $m_{t+1}^i = \frac{\beta u'(c_{t+1}^i)}{u'(c_t^i)}$. Show that the stochastic discount factors of consumer 1 and 2 are $m_{t+1}^1 = \beta + .25 \frac{\beta(1-\beta)}{(\gamma - c_t^1)}$ if $s_{t+1} = 0$, $\beta - .25 \frac{\beta(1-\beta)}{(\gamma - c_t^1)}$ if $s_{t+1} = 1$; $m_{t+1}^2 = \beta - .25 \frac{\beta(1-\beta)}{(\gamma - c_t^2)}$ if $s_{t+1} = 0$, $\beta + .25 \frac{\beta(1-\beta)}{(\gamma - c_t^2)}$ if $s_{t+1} = 1$. Are the stochastic discount factors of the two consumers equal?
- (f) Verify that $E_t m_{t+1}^1 = E_t m_{t+1}^2 = \beta$.

Solution:

- (a) Apply (2.12.9), $c_t = (1 - \beta) \left[\sum_{j \geq 0} \beta^j E_t y_{t+j} - b_t \right]$. With iid endowments, $E_t y_{t+j}^i =$

.75 for $j \geq 1$, so $\sum_j \beta^j E_t y_{t+j}^i = y_t^i + \frac{.75\beta}{1-\beta}$ and

$$c_t^i = (1 - \beta)(y_t^i - b_t^i) + .75\beta.$$

Substituting into the budget constraint $b_{t+1}^i = \beta^{-1}(c_t^i + b_t^i - y_t^i)$:

$$b_{t+1}^i = \beta^{-1}[(1-\beta)y_t^i - (1-\beta)b_t^i + .75\beta + b_t^i - y_t^i] = \beta^{-1}[\beta b_t^i - \beta y_t^i + .75\beta] = b_t^i + .75 - y_t^i.$$

Consumer 1: $s_t = 0 \Rightarrow y_t^1 = 1 \Rightarrow b_{t+1}^1 = b_t^1 - .25$; $s_t = 1 \Rightarrow y_t^1 = .5 \Rightarrow b_{t+1}^1 = b_t^1 + .25$. Consumer 2: $s_t = 0 \Rightarrow y_t^2 = .5 \Rightarrow b_{t+1}^2 = b_t^2 + .25$; $s_t = 1 \Rightarrow y_t^2 = 1 \Rightarrow b_{t+1}^2 = b_t^2 - .25$. ✓

- (b) $b_{t+1}^i - b_t^i = .75 - y_t^i$ is iid with mean zero: b_t^i is a random walk, so it has a unit root, and so does $c_t^i = (1 - \beta)(y_t^i - b_t^i) + .75\beta$. But

$$(1 - \beta)b_t^i + c_t^i = (1 - \beta)y_t^i + .75\beta$$

is iid, hence stationary: c_t^i, b_t^i are cointegrated with vector $[(1 - \beta) \ 1]$, as in (2.12.19).

- (c) $b_{t+1}^i = b_t^i + .75 - y_t^i$ is a function of (b_t^i, s_t) , known at t ; it does not depend on s_{t+1} or on anything dated $t + 1$. The payoff of the IOU is fixed when it is issued.
- (d) Summing the debt laws: $(b^1 + b^2)_{t+1} - (b^1 + b^2)_t = 1.5 - (y_t^1 + y_t^2) = 1.5 - 1.5 = 0$, so with $b_0^1 + b_0^2 = 0$, $b_t^1 + b_t^2 = 0$ for all t . Then $c_t^1 + c_t^2 = (1 - \beta)(1.5 - 0) + 1.5\beta = 1.5$. Aggregate consumption equals the aggregate endowment, which is constant. The economy is closed: whatever consumer 1 borrows, consumer 2 lends. The only asset is a one-period IOU between them at gross rate β^{-1} , and it lets them smooth each other's shocks.
- (e) $u'(c) = \gamma - c$, so $m_{t+1}^i = \beta^{\frac{\gamma - c_{t+1}^i}{\gamma - c_t^i}} = \beta - \beta^{\frac{c_{t+1}^i - c_t^i}{\gamma - c_t^i}}$. From part a and the debt law, $c_{t+1}^i - c_t^i = (1 - \beta)[(y_{t+1}^i - y_t^i) - (b_{t+1}^i - b_t^i)] = (1 - \beta)(y_{t+1}^i - .75)$. Consumer 1: $s_{t+1} = 0 \Rightarrow \Delta c^1 = +.25(1 - \beta)$; $s_{t+1} = 1 \Rightarrow \Delta c^1 = -.25(1 - \beta)$. Hence

$$m_{t+1}^1 = \begin{cases} \beta - .25\frac{\beta(1-\beta)}{\gamma - c_t^1}, & s_{t+1} = 0 \\ \beta + .25\frac{\beta(1-\beta)}{\gamma - c_t^1}, & s_{t+1} = 1 \end{cases} \quad m_{t+1}^2 = \begin{cases} \beta + .25\frac{\beta(1-\beta)}{\gamma - c_t^2}, & s_{t+1} = 0 \\ \beta - .25\frac{\beta(1-\beta)}{\gamma - c_t^2}, & s_{t+1} = 1 \end{cases}$$

These have the **opposite signs** to the printed statement (see the trap box). They are **not equal**: in each state one consumer's SDF lies above β and the other's below. With only a risk-free bond, markets are incomplete, and the consumers cannot trade state-contingent claims that would equate their marginal rates of substitution state by state.

- (f) $E_t m_{t+1}^1 = .5(\beta - d_t^1) + .5(\beta + d_t^1) = \beta$ with $d_t^1 = .25\beta(1 - \beta)/(\gamma - c_t^1)$; likewise for consumer 2. Both SDFs price the one asset that is traded, the risk-free bond, at the common price $\beta = R^{-1}$. This is the Euler equation $E_t u'(c_{t+1}) = u'(c_t)$ in SDF form.

Verification. `ls_ch02.ipynb`, § 2.27 ($\gamma = 3$, 300 periods): debt laws, $b^1 + b^2 = 0$, $c^1 + c^2 = 1.5$, the cointegrating residual, the SDF values (.94475, .95525) for consumer 1, and $E_t m = \beta$. ✓

Where this goes wrong

The statement of 2.27(e) attaches “+” to $s_{t+1} = 0$ for consumer 1. But $s_{t+1} = 0$ is consumer 1’s *high*-endowment state ($y^1 = 1$): his consumption rises, his marginal utility $\gamma - c$ falls, so $m_{t+1}^1 < \beta$. The signs as printed would be right only if the endowment labels were swapped. The answers to “are they equal?” (no) and to part f are unaffected.

Exercise 2.30 Phelps and Pollak (1968) Meet Howard

Question

A sequence of decision makers at dates $t = 0, 1, \dots$, chooses $\{x_{t+1}, u_t\}_{t=0}^{\infty}$ subject to $x_{t+1} = Ax_t + Bu_t$, $t \geq 0$ where A is an $n \times n$ matrix, B is an $n \times k$ matrix, u_t is a $k \times 1$ vector of time t “controls”, and x_0 is a given initial condition. Let $r(x_t, u_t) = -(x_t'Rx_t + u_t'Qu_t)$, where R and Q are positive definite matrices. A time t decision maker chooses u_t, x_{t+1} . A time t decision maker’s preferences are ordered by $r(x_t, u_t) + \delta \sum_{j=1}^{\infty} \beta^j r(x_{t+j}, u_{t+j})$ (0), where $\beta \in (0, 1)$ and $\delta \in (0, 1]$.

(a) Let $V(x)$ solve the following Bellman equation: $V(x) = r(x, -Fx) + \beta V((A - BF)x)$ (1). Please interpret $V(x)$; i.e., please complete the sentence “ $V(x)$ is the value of ...”.

(b) For a given matrix F , please guess a functional form for $V(x)$, then describe an algorithm for solving the functional equation (1) for $V(x)$. Please get as far as you can in computing $V(x)$.

(c) Consider the functional equation $U(x) = \max_u \{r(x, u) + \delta\beta V(Ax + Bu)\}$ (2), where $V(x)$ satisfies the Bellman equation (1). Further, let $U(x)$ be attained by $u = -Gx$, so that $U(x) = r(x, -Gx) + \delta\beta V((A - BG)x)$. Please interpret $U(x)$ as a value function.

(d) Please define a Markov perfect equilibrium for the sequence of problems solved by the sequence of decision makers who choose $\{u_t\}_{t=0}^{\infty}$.

(e) Please describe how to compute a Markov perfect equilibrium in this setting.

(f) Please compare your algorithm for computing a Markov perfect equilibrium with the Howard policy improvement algorithm.

(g) Let $\vec{a}_0 = \{a_t\}_{t=0}^{\infty}$. Define $\vec{a}_1 = \{a_t\}_{t=1}^{\infty}$ as the continuation of the sequence $\vec{a}_0$. Is a continuation of a Markov perfect equilibrium a Markov perfect equilibrium?

(h) Suppose instead that there is a dictator who at time 0 chooses $\{u_t\}_{t=0}^{\infty}$ to maximize the time $t = 0$ value of the criterion (0). Please write Bellman equations and tell how to solve them for an optimal plan for the time 0 dictator.

(i) Given x_1 , a time 1 dictator chooses $\{u_t\}_{t=1}^{\infty}$ to maximize utility function (0) for time t . Is a continuation of the time 0 dictator’s plan the time 1 dictator’s plan?

(j) Can you restrict $\delta \in (0, 1]$ so that the time 0 dictator’s plan equals the outcome of the Markov perfect equilibrium that you described above?

Solution:

(a) Iterating (1), $V(x) = \sum_{j \geq 0} \beta^j r(x_j, -Fx_j)$ with $x_{j+1} = (A - BF)x_j$, $x_0 = x$. $V(x)$ is the value of following the rule $u = -Fx$ forever from state x , discounted exponentially at β , without the extra δ . It is the continuation value a current self attaches, scaled by $\delta\beta$, to the path chosen by future selves who use F : $r(x_t, u_t) + \delta \sum_{j \geq 1} \beta^j r = r(x_t, u_t) + \delta\beta V(x_{t+1})$.

(b) Guess $V(x) = -x'P_Fx$. Substituting into (1): $-x'P_Fx = -x'(R + F'QF)x -$

$\beta x'(A - BF)'P_F(A - BF)x$ for all x , so

$$P_F = R + F'QF + \beta(A - BF)'P_F(A - BF),$$

a discrete Lyapunov equation, i.e. (2.4.25). Algorithms: iterate $P_{k+1} = R + F'QF + \beta(A - BF)'P_k(A - BF)$ from $P_0 = 0$ (converges iff $\sqrt{\beta}(A - BF)$ is stable); or doubling; or in closed form $\text{vec}(P_F) = [I - \beta(A - BF)' \otimes (A - BF)]^{-1} \text{vec}(R + F'QF)$.

- (c) With $V = -x'P_Fx$ the maximand in (2) is $-x'Rx - u'Qu - \delta\beta(Ax + Bu)'P_F(Ax + Bu)$. First-order condition: $-2Qu - 2\delta\beta B'P_F(Ax + Bu) = 0$, so

$$u = -Gx, \quad G = \delta\beta(Q + \delta\beta B'P_FB)^{-1}B'P_FA.$$

$U(x)$ is the value, to the current self at state x , of choosing today's control optimally given that all future selves will follow $u = -Fx$, evaluated with the current self's quasi-hyperbolic weights (1 today, $\delta\beta^j$ later). It is a one-shot deviation value, like (2.4.27).

- (d) A (stationary, linear) **Markov perfect equilibrium** is a rule F such that, when every future self uses $u = -Fx$, each current self's best response is also $u = -Fx$: $G(F) = F$, i.e.

$$F = \delta\beta(Q + \delta\beta B'P_FB)^{-1}B'P_FA, \quad P_F = R + F'QF + \beta(A - BF)'P_F(A - BF).$$

Strategies depend only on the payoff-relevant state x_t , and the rule is a best response at every date and every state.

- (e) Iterate on the pair: start from a stabilising F_0 (e.g. the $\delta = 1$ optimum); given F_j , solve the Lyapunov equation for P_{F_j} (part b); set $F_{j+1} = \delta\beta(Q + \delta\beta B'P_{F_j}B)^{-1}B'P_{F_j}A$; stop when $\max |F_{j+1} - F_j| < \varepsilon$. The limit satisfies both equations in (d).
- (f) It is Howard's algorithm (2.4.30) **with two discount factors**: the evaluation step uses β (future selves discount exponentially among themselves), the improvement step uses $\delta\beta$ (today's self discounts tomorrow more). With $\delta = 1$ it is (2.4.30). With $\delta < 1$ the iteration is a fixed-point search for a game equilibrium, not an optimisation, so Howard's monotone improvement and convergence guarantees do not carry over. In the notebook example it converges in 10 ($\delta = .8$) and 14 ($\delta = .6$) iterations.
- (g) Yes. The MPE rule depends only on the current state and the problem at $t = 1$ has the same structure as at $t = 0$, so the continuation $\{u_t = -Fx_t\}_{t \geq 1}$ from any x_1 is again an MPE. The equilibrium is subgame perfect by construction.
- (h) From date 1 on, the dictator's objective $\delta \sum_{j \geq 1} \beta^j r$ is proportional to an ordinary exponentially discounted problem, so the continuation plan solves

$$W(x) = \max_u \{r(x, u) + \beta W(Ax + Bu)\}, \quad W(x) = -x'P^*x,$$

$$P^* = R + \beta A' P^* A - \beta^2 A' P^* B (Q + \beta B' P^* B)^{-1} B' P^* A,$$

$$F^* = \beta (Q + \beta B' P^* B)^{-1} B' P^* A,$$

solved e.g. by Howard iterations (2.4.30). The date-0 control solves $J(x_0) = \max_{u_0} \{r(x_0, u_0) + \delta \beta W(Ax_0 + Bu_0)\}$, whose first-order condition gives $u_0 = -F_0 x_0$, $F_0 = \delta \beta (Q + \delta \beta B' P^* B)^{-1} B' P^* A$. The plan is $u_0 = -F_0 x_0$ and $u_t = -F^* x_t$ for $t \geq 1$.

- (i) No (for $\delta < 1$). The time-1 dictator applies the same reasoning shifted one period and picks $u_1 = -F_0 x_1$, not the $u_1 = -F^* x_1$ the time-0 dictator planned for him. Preferences (0) are time inconsistent: today's self discounts between $t+1$ and $t+2$ at β , but when $t+1$ arrives that same trade-off is discounted at $\delta \beta$.
- (j) Yes: $\delta = 1$. Then $F_0 = F^*$, the dictator's plan is time consistent, and the MPE fixed point is F^* itself (a single Howard step from F^* returns F^*). For $\delta < 1$ the three rules differ. Example with $A = \begin{bmatrix} 1.02 & .1 \\ 0 & .9 \end{bmatrix}$, $B = [1, .5]'$, $R = I$, $Q = 1$, $\beta = .95$:

δ	MPE F	dictator F_0 (date 0)	dictator F^* ($t \geq 1$)
1	[.580578, .209347]	[.580578, .209347]	[.580578, .209347]
.8	[.535287, .193215]	[.534341, .192674]	[.580578, .209347]
.6	[.477215, .173063]	[.471727, .170097]	[.580578, .209347]

Present bias makes every self control less aggressively (smaller feedback), and the MPE differs from both pieces of the dictator's plan.

Verification. `ls_ch02.ipynb`, §2.30: the MPE satisfies $G(F) = F$; at $\delta = 1$ it equals the `solve_discrete_are` optimum; at $\delta < 1$ it does not. ✓

Companion code

Notebook: `Resolucao/ljungqvist_sargent/ls_ch02.ipynb`.

It reproduces every numerical result above (Python: `numpy`, `scipy`, `matplotlib`), executed end to end with outputs saved. Each section checks its numbers against an independent route with `assert`.

Item	What it does
helpers	<code>doublej</code> (doubling for $X = AXA' + CC'$, works with the constant's unit root), <code>kalman_steady</code> (iterates (2.7.14) to its fixed point), <code>save_pdf</code> (pgf backend, lmodern + cmap, so figure text is Type 1 and copyable).
§ 2.1–2.3, 2.10–2.12, 2.14, 2.16–2.18	Markov-chain likelihoods, stationary distributions, absorption probabilities, simulations, lake-model MoM/ML and a Monte Carlo of the asymptotic covariance.
§ 2.4–2.6, 2.8, 2.9, 2.29	Companion forms, Lyapunov and doubling, prediction and discounting formulas, population regression vs. simulated OLS, Cagan equilibria, SDF pricing by Monte Carlo, Howard-style Kalman filter.
§ 2.7, 2.15	<code>bigshow3</code> counterpart; Slutsky spectra and peak locations.
§ 2.19–2.24, 2.26, 2.28	Kalman filters (scalar, time-varying G_t , cross-products checked by brute-force Gaussian conditioning), innovations representations, VAR inversion, Kolmogorov–Szegő check.
§ 2.13, 2.25, 2.27, 2.30	Howard iterations vs. <code>solve_discrete_are</code> , impulse responses vs. (2.12.18), the two classic examples, the two-consumer loans economy, Markov perfect equilibrium.
<code>fig/ls02_*.pdf</code>	The seven figures used in this file.